Value-Weighted Retention Targeting Can Concentrate Harm: Diagnostics and Safeguards for Online Gambling

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Abstract

Retention incentives in online gambling are often allocated by ranking players on estimated treatment effect times lifetime value. Under a fixed budget this value-weighted rule maximises operator value, but operator value and player harm can rise together. We show that the harm exposure of any budget-exhausting rule exceeds the population mean by the covariance between treatment and harm divided by the budget, and give sufficient conditions for value-weighted ranking to make it non-negative. In a simulation where harm rises with value, the estimated value-weighted rule raises mean harm among treated players from 0.421 to 0.476 at a 30% budget, and further at a 10% budget. Harm tracking value is not enough: in one design an accurate value-weighted rule falls below the population mean despite a strong harm-value correlation. A harm weight on the score lowers exposure to about what effect-only ranking reaches with no harm input. With estimated components the weight also raises operator value, but not through harm information: the gain turns into a cost with accurate components and is small or absent unless harm rises with value, consistent with trimming the noisy upper tail of the product score. A hard cap beats the harm weight only with accurate components, and its ceiling holds only for the measure it is given. In this design, with estimated effects, ranking on the effect alone beats value weighting for every learner at budgets up to 30%, significantly in all but one cell; removing effect-churn moderation reverses this. All parameters are illustrative.

Keywords: policy learning, budget-constrained targeting, gambling harm, uplift modelling, responsible gambling

1. Introduction

An online gambling operator has a retention incentive, such as a bonus or a free bet, and a budget that covers only a fraction of its active players. The operator must choose whom to reach. Practice ranks players by the product of two estimates: the incentive's effect on retention and the player's lifetime value (Haupt and Lessmann, 2022; Gubela and Lessmann, 2021; Hitsch and Misra, 2024). The rule is attractive because it points the budget at players who both respond and are worth keeping.

In gambling the second factor is not neutral. Revenue is concentrated in a small group of heavy players, and that group carries a large share of gambling problems (Fiedler et al., 2019; Tom et al., 2014). Higher spend is associated with worse financial and health outcomes (Muggleton et al., 2021). A rule that ranks on value therefore risks ranking on harm. Regulators treat the risk as real. In Great Britain, licensees must stop marketing

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and the uptake of new bonus offers once strong indicators of harm appear (Gambling Commission, 2022), and new rules on promotions were announced in March 2025 and took effect in January 2026 (Gambling Commission, 2025). An operator that ranks on value and audits harm afterwards may find that the rule has already pointed its budget at the players the regulator asks it to protect.

This paper studies that problem inside budget-constrained empirical welfare maximisation (Kitagawa and Tetenov, 2018; Athey and Wager, 2021). The operator chooses a rule that treats at most a fraction κ of players to maximise operator value, the expected retained value created by the incentive. Operator value is a different quantity from player harm, and the two can move together. With an outcome value attached to each player, the optimal rule ranks by effect times value, which is the standard threshold result applied to a value-weighted outcome rather than a new argument. Our contribution is to ask what that rule does to harm, how to correct it, and what survives when its components are estimated.

Findings. The first finding is an identity. For any rule that exhausts the budget, the mean harm of treated players exceeds the population mean by the covariance between treatment and harm divided by the budget. Proposition 2 gives sufficient conditions under which value-weighted ranking makes that covariance non-negative: harm rises with value among players with the same treatment effect, and ranking on the effect does not itself steer away from harm. In our main design, where harm rises with value, the estimated rule raises mean harm among treated players from 0.421 to 0.476 at $\kappa = 0.30$, a gap of +0.055 (95% confidence interval +0.051 to +0.058), and the gap is +0.152 at $\kappa = 0.10$. The conditions are sufficient, not necessary, and harm rising with value is not enough on its own. In two cases we report, the second condition fails and the gap is negative: with feasible-oracle components, ranking on the effect alone has a gap of -0.046, and in a decoupled design the feasible-oracle value-weighted rule has a gap of -0.014 although harm and value are strongly correlated. Concentration is therefore a property of how effect, value and harm move together, not of value weighting alone.

The second finding concerns the remedy. A harm weight on the score brings exposure to 0.414 at $\kappa = 0.30$, close to the 0.416 that ranking on the estimated effect alone reaches with no harm input, although at $\kappa = 0.10$ it stays above the population mean. With estimated components the harm weight also raises operator value, by 15.8%, but we do not attribute that gain to harm information. The gain is large only where harm rises with value. It is much smaller where harm falls steeply with value, absent at $\kappa = 0.30$ where harm falls weakly with value or is independent of it, and becomes a cost with feasible-oracle components. That pattern fits a regulariser: because the harm measure tracks value, the weight trims the upper tail of the noisy product $\hat{\tau}\hat{V}$, where true effects are small or negative (Section 6.3). The hard cap is the population optimum for a harm ceiling, but with estimated components it earns less than the harm weight, and it guarantees only the harm that its harm measure sees.

The third finding is that, in the main design, value weighting pays only when its components are accurate. With feasible-oracle effects and values it wins at every budget we test. With estimated effects, ranking on the effect alone yields more operator value than value weighting for every learner we test at budgets up to $\kappa = 0.30$, and the difference is significant in all but one learner cell. This is a statement about one design and not about the rule. Its boundary is the null scenario of Appendix E, without effect-churn moderation, where value weighting wins with the default T-learner, as it does in the misspecified-effect and asymmetric-moderation scenarios (Section 6.6). A feasible oracle

that sees only covariates, and not the realised effect and value, is the right benchmark for what estimation can recover; how far it falls short of the hindsight oracle reflects the noise variances we chose.

The findings change what an operator should do before deployment. The harm diagnostic is cheap: it needs only a harm measure and the treatment assignment, and it can be computed for any rule, including the rule already in use. The soft weight should be chosen for what it does to harm exposure, and an operator should not count on its welfare gain carrying over to data in which the effect does not fall in the value tail. With accurate components both harm-aware rules cost operator value, so in that setting safety is paid for in revenue. We return to these points in Section 7.

Scope. Every optimality result is conditional on operator value being multiplicatively separable in treatment effect, value and preference weight (Assumption 2 and Definition 1). The $\hat{\tau} \cdot \hat{V}$ heuristic already takes this form, but it is an assumption. Section 8 gives a plausible operator preference under which it fails. All simulation parameters are illustrative, and no operator data informed them.

Contributions. We claim no new optimality result for value weighting. The contributions are a diagnostic, two harm-aware rules evaluated with their limits, a scoped study of value weighting under estimation, and a benchmark.

1. **A harm-concentration diagnostic.** We give the identity that exposure minus the population mean equals the covariance of treatment and harm divided by the budget, and sufficient conditions for a non-negative gap under value-weighted ranking (Proposition 2). In the main simulation design the estimated rule concentrates harm more than the feasible oracle, the gap is largest at tight budgets, and we report two cases in which the conditions fail and the gap is negative although harm tracks value (Section 6.2).
2. **Harm-aware rules and what their value effects mean.** We state a soft harm weight (Corollary 1) and a hard harm cap, which is a special case of empirical welfare maximisation with linear constraints (Corollary 2). With estimated components the soft weight lowers exposure and raises operator value, but the pattern across the harm sweep and a decoupled-harm design indicates that the value gain comes from trimming the noisy value tail and not from harm information, and under a noisy harm measure both rules overshoot their targets (Sections 6.3 and 6.4).
3. **Value weighting under estimation, in one design.** With feasible-oracle effects and values, value weighting wins at every budget. With estimated effects, effect-only ranking wins for every learner at budgets up to $\kappa = 0.30$ (significantly in all but one learner cell), the crossover budget $\kappa^\star$ exists only for some learners, and learner choice matters: a tuned S-learner, trained on Y and multiplied by $\hat{V}$, outperforms the tuned T-learner (Sections 6.5 and 6.6). The null scenario reverses the comparison, which marks the scope of the result.
4. **Benchmarking and attribution.** We measure estimation cost against a feasible oracle that sees only covariates, and a Shapley split attributes almost all regret to the effect estimate (Sections 6.5 and 6.6).

Roadmap. Section 2 reviews related work. Section 3 fixes notation and assumptions. Section 4 states the operator's decision problem, the optimal rule, the regret representation and the cost of a preference weight. Section 5 gives the harm-concentration result and the harm-aware rules. Section 6 reports the simulation study. Section 7 sets out preconditions for responsible deployment. Sections 8 and 9 discuss scope and conclude.

2. Related work

2.1. Policy learning and constrained welfare maximisation

The optimal treatment rule literature assigns treatment to maximise a population welfare criterion. Manski (2004) derives minimax-regret treatment rules under ambiguity, and Manski and Tetenov (2007) studies rules for a planner who is averse to risk in aggregate welfare. Kitagawa and Tetenov (2018) formulate empirical welfare maximisation (EWM) over a restricted class of rules, and Athey and Wager (2021) give doubly robust policy learning with regret guarantees of order $n^{-1/2}$ under complexity restrictions on the class. Bhattacharya and Dupas (2012) characterise welfare-maximising assignment under a budget constraint and derive the ranked-threshold structure directly. The rate of Athey and Wager (2021) is conditional on the complexity restriction, so it need not describe flexible learners used in practice. Our optimal rule is the threshold result of these papers applied to a value-weighted outcome (Remark 3).

Several papers add constraints or criteria beyond mean welfare. Sun (2024) studies EWM when the budget cost of a policy must itself be estimated, which covers moment constraints of the kind our hard harm cap imposes. Kitagawa and Tetenov (2021) allow welfare criteria that trade efficiency against equality. Viviano and Bradic (2024) learn targeting rules under fairness constraints on a protected group. Our hard cap (Corollary 2) is a special case of this literature, and we claim no novelty for it. What is new here is the setting: the constrained quantity is the harm of the players an incentive reaches, and the question is how a value-weighted rule behaves against it. The constraint is also unusual in a second way, because it binds only on the harm measure the rule is fed, so measurement error in that measure carries through to the guarantee.

A second strand supplies the estimation and evaluation tools such rules need. Kallus (2018) proposes balanced policy evaluation and learning, Zhao et al. (2012) develop outcome-weighted learning of treatment rules, and Zhou et al. (2023) study offline policy learning with multiple actions. Chernozhukov et al. (2018) give inference for heterogeneous effects with machine learning, Imai and Li (2023) evaluate treatment rules experimentally under a budget on the treated share, and Yadlowsky et al. (2025) evaluate prioritisation rules by rank-weighted average treatment effects. Our evaluation is simulation-based and uses known effects, so it complements rather than uses these estimators.

2.2. Value-weighted and budgeted uplift targeting

Haupt and Lessmann (2022) study targeting under response-dependent costs and rank players by a value- and cost-weighted treatment effect, of which $\hat{\tau}_i \cdot V_i$ is the cost-homogeneous case. Gubela and Lessmann (2021) propose value-driven evaluation metrics for uplift models, and the profit-uplift literature (Gubela et al., 2020; Fernández-Loría and Provost, 2022; Verbeke et al., 2017) frames targeting as expected-value maximisation. Hitsch and Misra (2024) characterise the optimal policy as a ranking by expected incremental profit. Budgeted ranking by an uplift score also has a long applied lineage in the Qini and profit-uplift literature (Radcliffe, 2007; Gubela et al., 2019; Devriendt et al., 2022), and Devriendt et al. (2022) note that budgeted targeting depends only on the rank order of the score. We claim no novelty for value weighting or for budgeted ranking. The contribution here is to study what value weighting does to harm, and how its performance changes when effect and value are estimated. This prior work evaluates ranking rules on marketing outcomes such as conversion or profit, where a high-value customer is not also

a customer the firm has a duty to protect. In gambling that separation fails, which is why the value factor needs scrutiny that the marketing literature has not given it.

2.3. Learners for heterogeneous effects

The framework is agnostic to the estimator of the effect $\tau(x)$. We use the T-, S- and X-learners (Künzel et al., 2019), a doubly robust learner (Kennedy, 2023) and the R-loss (Nie and Wager, 2021) for tuning, and causal forests (Wager and Athey, 2018) are a further option. Devriendt et al. (2018) survey uplift methods and benchmark them on marketing data.

2.4. Gambling harm, incentives and regulation

Gambling revenue is concentrated. Fiedler et al. (2019) find that a small group of problem gamblers accounts for a large share of spending, and Tom et al. (2014) find that a small share of online gamblers accounts for most activity and most problems. Muggleton et al. (2021) link higher gambling spend in bank-transaction data to adverse financial, social and health outcomes. Harm is not confined to problem gamblers: Browne et al. (2016) show that it is spread across risk levels, and Wardle et al. (2019) frame gambling harm as a public-health cost. Harm measurement is contested and imperfect (Delfabbro and King, 2019), which is why we study a noisy harm measure. Behavioural markers in account data can flag risk (Braverman and Shaffer, 2012; Auer and Griffiths, 2023), and Delfabbro et al. (2025) validate such markers against the Problem Gambling Severity Index. We treat a harm score of this kind as an input and do not model how it is produced.

Inducements are a regulated concern. Hing et al. (2017) analyse the structural features of betting inducements for harm minimisation, and Newall et al. (2025) analyse free bets and play-through requirements and propose disclosure of their economic value. In Great Britain the Customer Interaction guidance requires licensees to stop marketing and the uptake of new bonus offers when strong harm indicators are identified (Gambling Commission, 2022). Licence condition SR code 5.1.1 requires incentive terms to be clear and fair (Gambling Commission, n.d.), and revisions announced on 26 March 2025, in force from 19 January 2026, cap wagering requirements at ten times and limit an incentive to one product type (Gambling Commission, 2025), within the wider reform set out in the Department for Digital, Culture, Media and Sport (2023) white paper. These rules constrain a value-weighted targeter directly, and they motivate treating harm as part of the targeting problem rather than as a filter applied afterwards.

Outside gambling, retention research shows that targeting by predicted risk is often ineffective and that responsiveness should drive allocation (Ascarza, 2018; Ascarza et al., 2018; Lemmens and Gupta, 2020). Taken together, the literature gives us three inputs: evidence that revenue and harm concentrate on the same players, regulatory rules that bind targeting on harm indicators, and a policy-learning toolkit that can express a harm constraint. We are not aware of work that asks whether value-weighted retention targeting concentrates harm in gambling, or that compares harm-aware rules under estimated components, which is the gap this paper addresses.

3. Setup and notation

Consider N individuals indexed by i, each with a pre-treatment covariate vector $X_i \in \mathcal{X} \subseteq \mathbb{R}^d$, a binary treatment $W_i \in \{0, 1\}$ (a retention bonus) and a binary outcome $Y_i \in \{0, 1\}$ (retention over the evaluation window). The potential outcomes are $Y_i(0)$ and $Y_i(1)$.

Assumption 1 (Causal identification)**.** (i) *Unconfoundedness:* $(Y_i(0), Y_i(1)) \perp W_i \mid X_i$. (ii) *Bounded overlap:* there is a constant $\underline{e} > 0$ such that $\underline{e} \le \mathbb{P}(W_i = 1 \mid X_i = x) \le 1 - \underline{e}$ for every $x \in \mathcal{X}$. (iii) *Stable unit treatment value assumption (SUTVA):* there is no interference between units. (iv) *Consistency:* $Y_i = Y_i(W_i)$.

Under Assumption 1 the conditional average treatment effect (CATE) is identified:

$$\tau(x) = \mathbb{E}[Y_i(1) - Y_i(0) \mid X_i = x]. \tag{1}$$

In a randomised experiment all four parts hold by design and $\tau(x)$ can be estimated from the observed data directly. In observational data, estimating $\tau(x)$ requires a propensity model or an estimator that absorbs confounding (Section 7).

Assumption 2 (Incremental operator value)**.** For each individual i, $V_i \ge 0$ is the operator value of retaining i over the evaluation horizon, and $R_i \in [0,1]$ and $H_i \in [0,1]$ are the risk covariate and the harm status. (i) *Pre-treatment:* V_i, R_i and H_i are determined before treatment and are not affected by it, so $V_i(1) = V_i(0) = V_i$, and likewise for R_i and H_i. (ii) *Retention is the only channel:* treating i changes operator value only through retention, so the incremental operator value of treating i is $\big(Y_i(1) - Y_i(0)\big)V_i$. The (homogeneous) cost of treatment is accounted for by the budget constraint.

The operator also chooses a measurable preference weight $g : [0,1] \to [0,\infty)$ on the risk covariate, with $g \equiv 1$ the neutral choice. Under Assumption 2, define the *score*

$$s(x) = \mathbb{E}\Big[\big(Y_i(1) - Y_i(0)\big)\, V_i\, g(R_i) \;\Big|\; X_i = x\Big]. \tag{2}$$

If the individual effect $Y_i(1) - Y_i(0)$ is independent of (V_i, R_i) given X_i, then $s(x) = \tau(x)\,\mathbb{E}[V_i\, g(R_i) \mid X_i = x]$, and if V_i and R_i are functions of X_i this is the product $\tau(x)\, V(x)\, g(R(x))$. Part (i) is what makes V_i and R_i usable as weights: a value or risk score that the treatment itself moves would introduce post-treatment bias. Part (ii) rules out channels such as a bonus changing how much a retained player later spends; Section 8 discusses response-dependent costs.

Remark 1 (Pre-treatment requirement). Lifetime value and churn risk in real systems are usually estimated from behavioural histories that earlier promotions shaped. Using such estimates as V_i or R_i violates Assumption 2(i). Compliance requires cohorts defined before any promotional activity, or auxiliary models restricted to features that pre-date treatment exposure. A feature-provenance audit is therefore a precondition for applying the rule on top of existing value and churn models.

Players differ in harm status H_i. Section 5 defines the conditional mean $\bar{H}(x)$, the operator's pre-treatment harm measure $\hat{H}_i$ and the harm exposure $\eta(\pi)$ of a rule. A treatment rule is a measurable $\pi : \mathcal{X} \to [0,1]$, deterministic when it takes values in $\{0,1\}$, and the budget set is

$$\Pi_\kappa = \Big\{\pi : \mathcal{X} \to [0,1] \;\Big|\; \mathbb{E}[\pi(X_i)] \le \kappa\Big\}, \qquad \kappa \in (0,1). \tag{3}$$

Estimated quantities carry a hat. Table 1 collects the symbols.

4. The operator's decision problem

The operator ranks players by the incremental operator value that treating them adds, subject to the budget. This section defines that objective, gives the optimal rule, represents the cost of estimation error and shows what a preference weight costs.

Table 1: Notation reference.

Symbol	Meaning	First used
$X_i \in \mathcal{X}$	pre-treatment covariate vector	§3
$W_i \in \{0,1\}$	treatment indicator (distinct from $\mathcal{W}(\pi)$)	§3
$Y_i \in \{0,1\}$	binary outcome (retention)	§3
$\tau(x)$	conditional average treatment effect	§3
$V_i \geq 0$	operator value of retaining i	§3
$R_i \in [0,1]$	risk covariate (churn probability)	§3
$H_i \in [0,1]$	player harm status	§3
g	operator preference weight on R_i	§3
$s(x)$	score $\mathbb{E}[(Y_i(1) - Y_i(0))V_i g(R_i) \mid X_i = x]$	§3
$\kappa \in (0,1)$	budget (fraction treatable)	§3
Π_κ	budget set of treatment rules	§3
$\mathcal{W}(\pi; g)$	operator welfare of rule π	§4
c^*, π^*	optimal threshold and rule; $\hat{\pi}$ an estimated rule	§4
$\mathcal{R}$	regret of an estimated rule	§4
$\bar{H}(x)$	conditional mean harm $\mathbb{E}[H_i \mid X_i = x]$	§5
$\hat{H}_i$, $\hat{h}(x)$	operator's harm measure and its conditional mean	§5
ω	soft harm weight, non-increasing in $\hat{h}$	§5
$g_{\min}$	floor of the preference weight g	Appendix B
$\eta(\pi)$, $\eta_{\max}$	harm exposure of a rule; ceiling on measured exposure	§5
λ, μ	multipliers on the budget and the harm constraint	§5

Definition 1 (Operator welfare)**.** For a measurable $g : [0,1] \to [0,\infty)$ and a treatment rule $\pi : \mathcal{X} \to [0,1]$ (deterministic rules take values in $\{0,1\}$), the operator welfare of π is

$$\mathcal{W}(\pi; g) = \mathbb{E}\Big[\pi(X_i)\,(Y_i(1) - Y_i(0))\,V_i\,g(R_i)\Big] = \mathbb{E}\Big[\pi(X_i)\,s(X_i)\Big], \tag{4}$$

where the second equality follows from iterated expectations because $\pi(X_i)$ is a function of X_i. We assume $\mathbb{E}|s(X_i)| < \infty$.

Proposition 1 (Optimal budget-constrained rule)**.** *Let Assumption 2 hold, let $\kappa \in (0,1)$, and suppose (a) $\mathbb{P}(s(X_i) > 0) \geq \kappa$, and (b) $\mathbb{P}(s(X_i) = c) = 0$ for every $c > 0$. Let $c^* = \inf\{c \geq 0 : \mathbb{P}(s(X_i) > c) \leq \kappa\}$. Then $c^* \geq 0$, $\mathbb{P}(s(X_i) > c^*) = \kappa$, and the rule*

$$\pi^*(x) = \mathbf{1}\{s(x) > c^*\} \tag{5}$$

maximises $\mathcal{W}(\pi; g)$ over all measurable $\pi : \mathcal{X} \to [0,1]$ with $\mathbb{E}[\pi(X_i)] \leq \kappa$, randomised rules included. Any other maximiser equals π^* almost surely. Equivalently, π^* treats the top-κ fraction of the population ranked by s; when $c^* > 0$, ties at the threshold occur with probability zero.

Proof sketch. For any feasible π,

$$\mathcal{W}(\pi^*; g) - \mathcal{W}(\pi; g) = \mathbb{E}\Big[(\pi^* - \pi)(s - c^*)\Big] + c^*\Big(\kappa - \mathbb{E}[\pi]\Big).$$

The integrand of the first term is non-negative pointwise: $\pi^* = 1 \geq \pi$ where $s > c^*$, and $\pi^* = 0 \leq \pi$ where $s < c^*$. The second term is non-negative because $c^* \geq 0$ and $\mathbb{E}[\pi] \leq \kappa$. Condition (a) gives $c^* \geq 0$ and condition (b) gives $\mathbb{P}(s > c^*) = \kappa$ and almost-sure uniqueness. No constraint qualification is needed; the full argument is in Appendix A. This is the threshold logic of Kitagawa and Tetenov (2018, Theorem 1) and Bhattacharya and Dupas (2012) applied to the score s. □

Remark 2 (Slack budget). If $\mathbb{P}(s(X_i) > 0) < \kappa$, the budget does not bind: every rule that treats $\{s > 0\}$, does not treat $\{s < 0\}$, and spends any remaining budget on $\{s = 0\}$ is optimal, so the optimum is not unique.

Remark 3 (Reduction to a transformed outcome). Let $\tilde{Y}_i = Y_i\, V_i\, g(R_i)$. By Assumption 2(i), $\tilde{Y}_i(w) = Y_i(w)\, V_i\, g(R_i)$, so the CATE of $\tilde{Y}$ is exactly $s(x)$ in (2), and $\mathcal{W}(\pi; g)$ is the unweighted EWM welfare of Kitagawa and Tetenov (2018) for $\tilde{Y}$. Proposition 1 is therefore their threshold result for a transformed outcome, not a new argument. The product form $\tau V g$ is the factorisation of s under the conditional-independence condition stated after Assumption 2; the decomposed rule estimates the factors separately, while a joint learner trained on $\tilde{Y}$ estimates s directly.

Setting $V_i = 1$ and $g \equiv 1$ recovers the welfare of Kitagawa and Tetenov (2018), and $g \equiv 1$ alone gives the value-weighted welfare behind the practice of Haupt and Lessmann (2022) and Gubela and Lessmann (2021). We call $\mathcal{W}$ *operator welfare* and reserve *player harm* for H_i: the results here concern what the operator maximises, and the two can point in different directions.

Remark 4 (Threshold and ordinal sufficiency). The threshold c^* is the shadow price of the budget: the minimum score at which a unit of budget is worth spending, and the operator value that one more unit of budget would buy. The rule depends on the scores only through their ranking, so an estimated score that ranks players as the true score does selects the same set. Rank preservation is a strong condition for a product of estimated components, because monotone distortions of the factors do not in general preserve the order of their product.

Lemma 1 (Regret representation)**.** *Under the conditions of Proposition 1, let $\hat{\pi} : \mathcal{X} \to \{0, 1\}$ be any deterministic rule with $\mathbb{E}[\hat{\pi}(X_i)] \le \kappa$, for example a top-$\kappa$ rule on an estimated score fitted on an independent sample. Its regret $\mathcal{R}(\hat{\pi}) = \mathcal{W}(\pi^*; g) - \mathcal{W}(\hat{\pi}; g)$ satisfies*

$$\mathcal{R}(\hat{\pi}) = \mathbb{E}\Big[\, |s(X_i) - c^*|\, \mathbf{1}\{\hat{\pi}(X_i) \neq \pi^*(X_i)\}\Big] + c^*\Big(\kappa - \mathbb{E}[\hat{\pi}(X_i)]\Big), \tag{6}$$

and the second term vanishes when $\hat{\pi}$ exhausts the budget.

Proof. Apply the decomposition in the proof of Proposition 1 with $\pi = \hat{\pi}$. For binary rules, $(\pi^* - \hat{\pi})(s - c^*)$ equals $s - c^* > 0$ where $\pi^* = 1, \hat{\pi} = 0$, equals $c^* - s \ge 0$ where $\pi^* = 0, \hat{\pi} = 1$, and is zero where the rules agree, which is $|s - c^*|\, \mathbf{1}\{\hat{\pi} \neq \pi^*\}$. □

Regret is driven only by individuals the two rules disagree on, weighted by their distance from the threshold. This is the margin representation behind fast rates for plug-in classifiers (Audibert and Tsybakov, 2007) and the threshold structure of Kitagawa and Tetenov (2018); we use it descriptively and derive no rates.

Remark 5 (Order-free attribution of regret). For a decomposed rule ranking by $\hat{\tau}\hat{V}$, let $\mathcal{R}(a, b)$ be the regret of the top-κ rule that ranks by $a \cdot b$, measured against the feasible oracle that ranks by $\bar{\tau}\bar{V}$ with $\bar{\tau}(x) = \mathbb{E}[\tau_i \mid X_i = x]$ and $\bar{V}(x) = \mathbb{E}[V_i \mid X_i = x]$, so that $\mathcal{R}(\bar{\tau}, \bar{V}) = 0$. Replacing one component at a time attributes regret in an order-dependent way. The two-player Shapley values average over both orders:

$$\phi_\tau = \tfrac{1}{2}\Big[\mathcal{R}(\hat{\tau}, \bar{V}) + \mathcal{R}(\hat{\tau}, \hat{V}) - \mathcal{R}(\bar{\tau}, \hat{V})\Big], \qquad \phi_V = \tfrac{1}{2}\Big[\mathcal{R}(\bar{\tau}, \hat{V}) + \mathcal{R}(\hat{\tau}, \hat{V}) - \mathcal{R}(\hat{\tau}, \bar{V})\Big],$$

and $\phi_\tau + \phi_V = \mathcal{R}(\hat{\tau}, \hat{V})$. A share $\phi_\tau / \mathcal{R}$ above one means that, averaged over orders, swapping in $\hat{V}$ reduces regret, so the value estimate partly offsets error in $\hat{\tau}$.

Observation 1 (Preference weights and plug-in calibration)**.** *Let π_g^* maximise $\mathcal{W}(\pi; g)$ over the budget set and let $\pi_{\mathbf{1}}^*$ maximise $\mathcal{W}(\pi; \mathbf{1})$. Then $\mathcal{W}(\pi_g^*; \mathbf{1}) \leq \mathcal{W}(\pi_{\mathbf{1}}^*; \mathbf{1})$, because π_g^* is feasible for the problem that $\pi_{\mathbf{1}}^*$ solves; under the conditions of Proposition 1 for $g \equiv 1$, equality holds if and only if $\pi_g^* = \pi_{\mathbf{1}}^*$ almost surely. The same argument applies on any sample. Suppose g is chosen from a finite candidate set containing the identity by maximising the plug-in welfare $\widehat{\mathcal{W}}_n(\pi) = n^{-1} \sum_{i=1}^n \pi(X_i)\, \hat{\tau}(X_i) \hat{V}(X_i)$ on a validation sample, where each candidate rule π_g treats the $\lfloor \kappa n \rfloor$ units with the largest $\hat{\tau} \hat{V} g(\hat{R})$. The identity candidate treats the $\lfloor \kappa n \rfloor$ units with the largest $\hat{\tau} \hat{V}$, which maximises $\widehat{\mathcal{W}}_n$ over all sets of that size, so the identity attains the maximum on every sample. Plug-in calibration therefore selects the identity (or a candidate with the same plug-in welfare) by construction, and its selection rate carries no evidence about whether g helps. Evidence must come from an objective that is not maximised by construction, such as a doubly robust, augmented inverse-propensity-weighted (AIPW) estimate of $\mathcal{W}(\pi; \mathbf{1})$.*

Remark 6 (Decomposed and joint scores). A decomposed rule estimates the factors of s separately, so each model can be audited on its own. A joint learner trained on $\tilde{Y}$ estimates s directly and need not form the product. The S-learner of Section 6 is not of this kind: it is trained on Y and multiplied by $\hat{V}$.

5. Harm concentration and the harm-aware rule

In gambling, a player's value to the operator is large partly because the player gambles in large volumes, and some of those volumes are harmful. Value weighting can therefore pull incentives toward harmed players although nothing in the score measures harm. Whether it does is an empirical question about the joint distribution of effect, value and harm, and Proposition 2 gives sufficient conditions for it. The exclusion sets, fail-closed defaults and regulatory context that surround any such rule are in Section 7.

Let $H_i \in [0, 1]$ denote true harm, $\bar{H}(x) = \mathbb{E}[H_i \mid X_i = x]$, and for a rule with $\mathbb{E}[\pi(X_i)] > 0$ define the harm exposure $\eta(\pi) = \mathbb{E}[H_i \mid \pi(X_i) = 1]$, the mean harm among treated players.

Proposition 2 (Harm concentration)**.** *(i) For every deterministic rule π with $\mathbb{E}[\pi(X_i)] = \kappa$,*

$$\eta(\pi) - \mathbb{E}[H_i] = \frac{\operatorname{Cov}\Big(\pi(X_i), \bar{H}(X_i)\Big)}{\kappa}. \tag{7}$$

(ii) Let $\pi(x) = \mathbf{1}\{a(x)\, b(x) > c\}$ with $b \geq 0$, $c \geq 0$ and $\mathbb{E}[\pi(X_i)] = \kappa$, where $(a, b) = (\tau, \bar{V})$ for the oracle rule or $(\hat{\tau}, \hat{V})$ for an estimated rule. Suppose (A) $\mathbb{E}[\bar{H}(X_i) \mid a(X_i), b(X_i)] = \psi(a(X_i), b(X_i))$ for a function ψ that is non-decreasing in its second argument for every value of the first, and (B) $\operatorname{Cov}\Big(\mathbb{P}(\pi(X_i) = 1 \mid a(X_i)),\ \mathbb{E}[\bar{H}(X_i) \mid a(X_i)]\Big) \geq 0$, which holds in particular when $\mathbb{E}[\bar{H}(X_i) \mid a(X_i)]$ does not depend on $a(X_i)$. Then $\eta(\pi) \geq \mathbb{E}[H_i]$. The inequality is strict if, in addition, $\psi(u, \cdot)$ is strictly increasing and $0 < \mathbb{P}(\pi(X_i) = 1 \mid a(X_i) = u) < 1$ for a set of $u > 0$ of positive probability. Reversing the monotonicity in (A) and the sign in (B) reverses the conclusion.

In words: whenever harm rises with value among players with the same treatment effect, and treatment-effect ranking does not itself steer away from harm, ranking by effect times value concentrates harm among the treated. The conditions are sufficient and not necessary. Condition (A) alone does not settle the sign: when (B) fails, because players with large effects have low harm, the second term of the law-of-total-covariance

decomposition in the proof is negative and can outweigh the first, and Section 6.2 reports two such cases. The conditions concern the ranking components, so they can hold for the estimated rule and fail for the oracle rule, or the reverse. No such association is implied by the targeting problem itself, so concentration is a property of the joint distribution of effect, value and harm, not of the rule; Section 6.2 sweeps the harm-value alignment c_{harm} from -1 to 1.

Proof sketch. (i) is iterated expectations; (ii) applies the law of total covariance conditional on $a(X)$ and Chebyshev's association inequality within each level of $a(X)$; see Appendix A.

Let $\hat{H}_i \in [0,1]$ be the operator's harm measure, observed before treatment and available to the rule, and write $\hat{h}(x) = \mathbb{E}[\hat{H}_i \mid X_i = x]$, which equals $\hat{H}_i$ when the rule conditions on $\hat{H}_i$. In the simulation $\hat{H}_i$ enters only the harm-aware rules; the effect, value and churn models use the four covariates of Section 6.1.

Corollary 1 (Soft harm weight)**.** *Let $\omega : [0,1] \to [0,1]$ be measurable and non-increasing. Suppose the score $s^\dagger(x) = s(x)\,\omega(\hat{h}(x))$ satisfies conditions (a) and (b) of Proposition 1 in place of s, and let $c^\dagger = \inf\{c \geq 0 : \mathbb{P}(s^\dagger(X_i) > c) \leq \kappa\}$. Then the rule $\mathbf{1}\{s^\dagger(x) > c^\dagger\}$ maximises $\mathbb{E}[\pi(X_i)\, s^\dagger(X_i)]$ over the budget set. By the argument of Observation 1, its operator welfare $\mathcal{W}(\cdot; g)$ is at most that of the rule without the harm weight.*

The weight is soft: a large enough $s(x)$ clears the threshold whatever $\omega(\hat{h}(x))$ is, so Corollary 1 bounds no exposure. Capping value at a constant $v_{\max}$ is a multiplicative weight of the same form, $\min(1, v_{\max}/V)$, applied to value instead of harm. When a ceiling is required, it must be a constraint. For a ceiling $\eta_{\max}$ on measured exposure, consider

$$\max_{\pi:\mathcal{X}\to[0,1]} \mathbb{E}[\pi(X_i)\, s(X_i)] \quad \text{s.t.} \quad \mathbb{E}[\pi(X_i)] \leq \kappa, \qquad \mathbb{E}[\pi(X_i)\, \hat{h}(X_i)] \leq \eta_{\max}\, \kappa. \tag{8}$$

Corollary 2 (Hard harm cap)**.** *Suppose there are multipliers $\lambda^* \geq 0$ and $\mu^* \geq 0$ such that $\pi^\ddagger(x) = \mathbf{1}\{s(x) - \mu^* \hat{h}(x) > \lambda^*\}$ satisfies both constraints of (8), with $\lambda^*\Big(\kappa - \mathbb{E}[\pi^\ddagger]\Big) = 0$ and $\mu^*\Big(\eta_{\max}\kappa - \mathbb{E}[\pi^\ddagger \hat{h}]\Big) = 0$. Then $\pi^\ddagger$ solves (8), and if $\mathbb{P}\Big(s(X_i) - \mu^* \hat{h}(X_i) = \lambda^*\Big) = 0$ every other solution equals $\pi^\ddagger$ almost surely. In particular, $\pi^\ddagger$ attains at least the operator welfare of any rule, including a soft-weight rule, whose measured exposure is at most $\eta_{\max}$ within the same budget. The bound $\mathbb{E}[\hat{h}(X_i) \mid \pi^\ddagger = 1] \leq \eta_{\max}$ needs a binding budget, $\mathbb{E}[\pi^\ddagger] = \kappa$, which holds in particular when $\lambda^* > 0$; without it only the slack form $\mathbb{E}[\hat{h}(X_i) \mid \pi^\ddagger = 1] \leq \eta_{\max}\,\kappa / \mathbb{E}[\pi^\ddagger]$ follows.*

The proof is the Lagrangian argument of Proposition 1 with a second multiplier; it is in Appendix A.

We take the existence of (λ^*, μ^*) as a hypothesis; for given multipliers it is a set of moment conditions that can be checked directly. In the simulation the budget is exhausted by top-κ selection and μ^* is the smallest value for which the measured-harm constraint holds, found by one-dimensional search.

Relation to constrained empirical welfare maximisation. Problem (8) is empirical welfare maximisation with two linear moment constraints on the rule, applied to the transformed outcome of Remark 3. Bhattacharya and Dupas (2012) derive the threshold structure of welfare-maximising assignment under a budget constraint, and the Lagrangian argument above extends it to a second linear constraint; Sun (2024) studies empirical welfare

maximisation when such constraints must themselves be estimated, and Kitagawa and Tetenov (2021) and Viviano and Bradic (2024) add equality and fairness criteria to the targeting objective. Corollary 2 is a special case of this literature, and we claim no novelty for it. What the gambling application adds is that the constrained quantity is a harm measure, which is noisy, so the guarantee needs the qualification in the next remark.

Remark 7 (The ceiling binds on measured harm). When the budget binds ($\mathbb{E}[\pi^\ddagger] = \kappa$, in particular when $\lambda^* > 0$), Corollary 2 guarantees $\mathbb{E}[\hat{H}_i \mid \pi^\ddagger = 1] \leq \eta_{\max}$, not $\eta(\pi^\ddagger) \leq \eta_{\max}$; when it does not bind, the guarantee is the slack form $\eta_{\max}\,\kappa/\mathbb{E}[\pi^\ddagger]$. The difference is the overshoot $\eta(\pi^\ddagger) - \mathbb{E}[\hat{H}_i \mid \pi^\ddagger = 1] = \mathbb{E}[H_i - \hat{H}_i \mid \pi^\ddagger = 1]$. Because the penalty favours players with low $\hat{H}_i$, the treated set over-represents players whose measurement error is negative, so the overshoot tends to be positive when $\hat{H}$ is noisy, even if the error has mean zero in the population. The same selection affects the soft weight, and the dominance statement of Corollary 2 holds for the true score s; with estimated scores it need not hold (Section 6.3).

6. Simulation study

The simulation asks three questions. Does value-weighted ranking concentrate harm, and when? What does a harm-aware rule cost the operator, and what does it guarantee? How do the budget and the learner that estimates the effect shape the value of value weighting? Because we generate the population, we know the true effect, value, churn probability and harm of every player, and every rule is scored on those true quantities. Section 6.1 gives the design, Section 6.2 the harm results, Section 6.3 the harm-aware rules, Section 6.4 a decoupled-harm test, Section 6.5 operator value and the budget diagnostic, and Section 6.6 the choice of learner. Appendix C to Appendix E hold calibration, the harm sweep and additional scenarios.

6.1. Design

Instantiation. The treatment W_i is a binary retention bonus, the outcome Y_i is binary retention over the evaluation window, the operator value V_i is lifetime value (LTV), and the risk covariate R_i is the churn probability, written p_i below. The window is an operator choice, and cross-window effects are outside the framework. The design reflects the higher-frequency, variable-stake end of online gaming, with repeated sessions, log-normal weekly spend and a multi-session login cadence. Lottery products have a slower cadence and a less extreme-tailed value distribution. The heavy-tailed value scenario of Appendix E brackets the tail, and we expect a lower-tail vertical to show a smaller value-weighting penalty.

Population. Four pre-treatment covariates are drawn independently of one another: account_age $\sim \mathcal{N}(24, 8)$ clipped to $[1, 60]$, weekly_spend $\sim \exp(\mathcal{N}(3.5, 0.8))$ clipped to $[5, 500]$, login_freq $\sim$ Poisson(4) clipped to $[1, 15]$ and bonus_history $\sim \mathcal{N}(3, 1.5)$ clipped to $[0, 10]$, abbreviated below to age, spend, login and bonus. Throughout, $\mathcal{N}(\mu, \sigma)$ denotes a normal distribution with mean μ and standard deviation σ. Let $z_\cdot$ denote a feature standardised with moments from a sample of 2,000,000 draws, so that no replication standardises on its own data. Churn probability, value, effect and baseline

retention are

$$\begin{aligned}
\operatorname{logit}(p_i) &= -1.0 + 0.04\,(30 - \text{age}_i) - 0.25\,\text{login}_i + 0.01\,(40 - \text{spend}_i) + 0.1\,z_{\text{bonus},i} + \varepsilon_i^{(p)}, \\
&\quad \varepsilon_i^{(p)} \sim \mathcal{N}(0, 0.4), && (9) \\
V_i &= 10\,\text{spend}_i + 40\,\text{login}_i + 5\,\text{age}_i + \varepsilon_i^{(V)}, \quad \varepsilon_i^{(V)} \sim \mathcal{N}(0, 80), && (10) \\
\tau_i &= \Big(0.06 + 0.10\,z_{\text{login},i} - 0.02\,z_{\text{login},i}^2 - 0.04\,z_{\text{spend},i}\Big)\,4p_i(1 - p_i) + \varepsilon_i^{(\tau)}, \\
&\quad \varepsilon_i^{(\tau)} \sim \mathcal{N}(0, 0.03), && (11) \\
b_i &= 0.6 + 0.15\,z_{\text{login},i} + 0.05\,z_{\text{spend},i} - 0.08\,z_{\text{age},i} + \varepsilon_i^{(b)}, \quad \varepsilon_i^{(b)} \sim \mathcal{N}(0, 0.05), && (12)
\end{aligned}$$

with p_i clipped to $[0.01, 0.99]$, V_i to $[50, 8{,}000]$, τ_i to $[-0.20, 0.40]$ and b_i to $[0.05, 0.95]$. Outcomes are $Y_i(0) \sim \text{Bernoulli}(b_i)$ and $Y_i(1) \sim \text{Bernoulli}\Big((b_i + \tau_i)_{[0.01,0.99]}\Big)$, drawn independently, which does not affect welfare because welfare is scored on τ_i. The effect is small on average and negative for many players: most churn probabilities are well below 0.5, so the moderation $4p(1-p)$ shrinks the base effect, and the effect noise is large relative to its mean. The effect also falls with spend, directly and through the moderation, because churn probability falls with spend. The moderation $4p(1-p)$ is symmetric, which is a simulation convenience: Ascarza (2018) documents faster attenuation at high churn, and Appendix E reruns the study with an asymmetric moderation. Treatment is a fair coin, $W_i \sim \text{Bernoulli}(0.5)$, so overlap (Assumption 1(ii)) holds with $\underline{e} = 0.5$ and the study does not probe low-propensity regimes. Welfare is scored on the nominal $\tau_i V_i$, which exceeds the clip-adjusted welfare of CATE $\times$ LTV by +787; on clip-adjusted welfare effect-only ranking still beats CATE $\times$ LTV at $\kappa = 0.30$ (Appendix E).

Harm. True harm is $H_i = \text{sigmoid}\Big(-0.3 + \sigma_L\,(c_{\text{harm}}\,L_i + \sqrt{1 - c_{\text{harm}}^2}\,Z_i) + e_i\Big)$, where $L_i = (1.1\,z_{\text{spend},i} + 0.4\,z_{\text{login},i})/\sigma_L$ is the standardised spend-login index, $\sigma_L = \sqrt{1.1^2 + 0.4^2}$, $Z_i \sim \mathcal{N}(0,1)$ and $e_i \sim \mathcal{N}(0, 0.5)$. The alignment $c_{\text{harm}} \in \{-1, -0.5, 0, 0.5, 1\}$ sets whether harm rises with value ($c_{\text{harm}} > 0$), is independent of it (0) or falls with it (< 0), and the variance of the harm logit does not depend on c_{harm}. The operator's measure is $\hat{H}_i = \text{sigmoid}\Big(\text{logit}(H_i) + \sigma_h\,\xi_i\Big)$ with $\xi_i \sim \mathcal{N}(0,1)$ and $\sigma_h \in \{0, 0.5, 1\}$. It is a noisy oracle built from true harm, not an estimator: no feature-based model could recover the Z_i and e_i components, so even $\sigma_h = 0$ is more favourable than any real measure. The harm sweep runs at $\kappa = 0.10$ and 0.30.

Estimation. Each replication splits the population 70/30 into training and test sets, fits every model on the training set and scores rules on the test set. The default T-learner fits one gradient-boosted classifier per arm (depth 3, 100 trees, learning rate 0.1) and differences the predicted probabilities. The S-learner fits one depth-4 gradient-boosted regressor with treatment as a feature, the X-learner follows Künzel et al. (2019), and the DR-learner regresses the cross-fitted doubly robust pseudo-outcome on X with the known propensity 0.5. Each learner also has a tuned version, chosen by cross-fitted R-loss over a grid of 80 depth, tree-count and learning-rate configurations in 5 pilot replications whose seeds are disjoint from the evaluation seeds (Appendix E). The default value model is a deliberately wrong proxy: a gradient-boosted regressor trained on $12\,\text{spend}^{0.9} + 35\,\text{login} + \varepsilon$ with $\varepsilon \sim \mathcal{N}(0, 200)$, which drops age. The alternative is a well-specified model trained on the true form of Equation (10). The churn model classifies control-arm non-retention, a proxy for p_i rather than p_i itself.

Rules. The strategies are Random; CATE only ($\hat{\tau}$); CATE $\times$ LTV ($\hat{\tau}\hat{V}$); and the S-learner ranking $\hat{\tau}_S\hat{V}$, which borrows $\hat{V}$. As a secondary comparison, two churn-weighted rules test the preference weight g of Section 4; they are not the rules this paper proposes, and they do not use harm. Churn-weighted (est.) multiplies $\hat{\tau}\hat{V}$ by $g(\hat{p};\theta_0)$ at the default $\theta_0 = (0.05, 0.40, 0.60, 0.90, 0.20)$ of Appendix B. Churn-weighted (cal.) chooses θ on a validation fold, by an AIPW objective or by the plug-in objective of Observation 1, from 22 configurations, one of which is a near-identity configuration that is equivalent to the identity because $\hat{p} \geq 0.01$. The harm-aware rules use $\hat{H}$. The soft harm weight ranks by $\hat{\tau}\hat{V}\,\omega(\hat{H})$ with ω falling linearly from 1 to 0 on $[0.6, 0.9]$. The hard harm cap ranks by $\hat{\tau}\hat{V} - \mu\hat{H}$, with μ the smallest value for which mean $\hat{H}$ among its treated does not exceed a ceiling, and the ceiling equals the soft weight's mean $\hat{H}$ among its treated, so the two rules have equal measured exposure. Each of the three harm-relevant rules (harm-blind, soft weight, hard cap) also runs with feasible-oracle components in place of $\hat{\tau}\hat{V}$, and CATE-only ranking is scored for harm exposure with estimated and feasible-oracle effects.

Oracles. The hindsight oracle ranks by the realised $\tau_i V_i$, which no rule can observe. The feasible oracle ranks by $\mathbb{E}[\tau \mid X]\,\mathbb{E}[V \mid X]$, which equals $\mathbb{E}[\tau V \mid X]$ because the effect and value noises are independent, so by Proposition 1 it is the best rule that uses covariates alone. Both use $g \equiv 1$ unless a table says θ_0. Estimation cost is measured against the feasible oracle.

Replications and inference. We run 200 replications of 10,000 players with seeds $42 + 7r$. Operator welfare is $\mathcal{W}(\pi; \mathbf{1}) = \sum_{i \text{ treated}} \tau_i V_i$ on the test set, and the reference budget is $\kappa = 0.30$; Section 6.5 varies κ over $\{0.10, 0.20, 0.30, 0.50\}$. We compare rules with paired t-tests across replications and report differences with 95% confidence intervals and Cohen's d of the paired differences; p-values appear in tables. Holm corrections run within named families and never across them. The families are the eight paired tests of each harm cell, the learner comparisons at each budget, the sixteen effect-only comparisons at each budget (a separate family), the budgets of each effect-source and value-source cell of the crossover analysis (across the whole κ grid), the two calibrated-rule comparisons at each budget of Table 6, the three non-degenerate calibrated-versus-identity comparisons of Table C.1, and the comparisons of each robustness scenario. We define $\kappa^\star$ as the smallest κ on a 0.025 grid at which $\hat{\tau}\hat{V}$ beats $\hat{\tau}$ alone and keeps beating it at every larger grid budget, a stable up-crossing. Its confidence interval is a percentile bootstrap over replications, censored at the grid ends, and we report how many of the 2,000 draws fall below, inside or above the grid.

6.2. Harm concentration

Table 2 sweeps the alignment c_{harm} and reports exposure minus the population mean, with 95% confidence intervals. At $c_{\text{harm}} = 1$ the estimated $\hat{\tau}\hat{V}$ rule treats players with mean harm 0.476 against a population mean of 0.421, a gap of +0.055 [+0.051, +0.058]. With feasible-oracle components the gap is +0.033 [+0.032, +0.034], smaller, so estimating the components concentrates harm more than the best covariate-only rule does.

In the main design the sign of the harm-blind gap follows c_{harm}. It is negative where harm falls with value, indistinguishable from zero where harm is independent of value, and positive where harm rises with value, consistent with Proposition 2, whose monotonicity condition describes harm rising with value. The sample correlation of harm and value runs from -0.75 at $c_{\text{harm}} = -1$ through +0.00 to +0.80 at $c_{\text{harm}} = 1$. The population

Table 2: Harm exposure of the treated minus the population mean ($\kappa = 0.30$, 95% confidence intervals in brackets). Panel A uses estimated components and panel B feasible-oracle components. The population mean is that of the row's c_{harm}. The harm-blind gap does not vary with σ_h because that rule never reads $\hat{H}$ and exposure is scored on true harm.

c_{harm}	σ_h	Harm-blind	Soft weight	Hard cap
Panel A: estimated components				
−1.0	0.0	−0.056 [−0.059, −0.053]	−0.076 [−0.079, −0.073]	−0.076 [−0.079, −0.073]
−1.0	0.5	−0.056 [−0.059, −0.053]	−0.079 [−0.082, −0.076]	−0.081 [−0.084, −0.078]
−1.0	1.0	−0.056 [−0.059, −0.053]	−0.080 [−0.083, −0.077]	−0.084 [−0.087, −0.081]
−0.5	0.0	−0.027 [−0.029, −0.025]	−0.079 [−0.080, −0.077]	−0.079 [−0.080, −0.077]
−0.5	0.5	−0.027 [−0.029, −0.025]	−0.079 [−0.081, −0.078]	−0.080 [−0.082, −0.078]
−0.5	1.0	−0.027 [−0.029, −0.025]	−0.079 [−0.081, −0.077]	−0.080 [−0.082, −0.078]
0.0	0.0	+0.000 [−0.001, 0.001]	−0.064 [−0.065, −0.062]	−0.064 [−0.065, −0.063]
0.0	0.5	+0.000 [−0.001, 0.001]	−0.065 [−0.066, −0.063]	−0.064 [−0.066, −0.063]
0.0	1.0	+0.000 [−0.001, 0.001]	−0.065 [−0.066, −0.063]	−0.064 [−0.065, −0.063]
0.5	0.0	+0.027 [0.025, 0.028]	−0.043 [−0.045, −0.042]	−0.043 [−0.045, −0.042]
0.5	0.5	+0.027 [0.025, 0.028]	−0.045 [−0.046, −0.043]	−0.043 [−0.044, −0.041]
0.5	1.0	+0.027 [0.025, 0.028]	−0.046 [−0.047, −0.044]	−0.041 [−0.043, −0.040]
1.0	0.0	+0.055 [0.051, 0.058]	−0.008 [−0.010, −0.005]	−0.008 [−0.010, −0.005]
1.0	0.5	+0.055 [0.051, 0.058]	−0.011 [−0.013, −0.008]	−0.003 [−0.005, 0.000]
1.0	1.0	+0.055 [0.051, 0.058]	−0.015 [−0.017, −0.012]	+0.002 [0.000, 0.005]
Panel B: feasible-oracle components				
−1.0	0.0	−0.044 [−0.045, −0.043]	−0.065 [−0.066, −0.064]	−0.065 [−0.066, −0.064]
−1.0	0.5	−0.044 [−0.045, −0.043]	−0.069 [−0.070, −0.068]	−0.073 [−0.074, −0.072]
−1.0	1.0	−0.044 [−0.045, −0.043]	−0.069 [−0.070, −0.068]	−0.078 [−0.079, −0.077]
−0.5	0.0	−0.015 [−0.016, −0.014]	−0.085 [−0.086, −0.085]	−0.086 [−0.087, −0.085]
−0.5	0.5	−0.015 [−0.016, −0.014]	−0.085 [−0.086, −0.084]	−0.086 [−0.087, −0.085]
−0.5	1.0	−0.015 [−0.016, −0.014]	−0.082 [−0.083, −0.081]	−0.084 [−0.085, −0.083]
0.0	0.0	+0.000 [−0.001, 0.001]	−0.088 [−0.088, −0.087]	−0.088 [−0.089, −0.087]
0.0	0.5	+0.000 [−0.001, 0.001]	−0.087 [−0.088, −0.086]	−0.087 [−0.088, −0.086]
0.0	1.0	+0.000 [−0.001, 0.001]	−0.083 [−0.084, −0.082]	−0.084 [−0.085, −0.083]
0.5	0.0	+0.011 [0.011, 0.012]	−0.070 [−0.071, −0.070]	−0.071 [−0.071, −0.070]
0.5	0.5	+0.011 [0.011, 0.012]	−0.070 [−0.071, −0.069]	−0.070 [−0.071, −0.070]
0.5	1.0	+0.011 [0.011, 0.012]	−0.068 [−0.069, −0.067]	−0.068 [−0.069, −0.068]
1.0	0.0	+0.033 [0.032, 0.034]	−0.012 [−0.013, −0.011]	−0.012 [−0.013, −0.011]
1.0	0.5	+0.033 [0.032, 0.034]	−0.016 [−0.017, −0.015]	−0.015 [−0.015, −0.014]
1.0	1.0	+0.033 [0.032, 0.034]	−0.019 [−0.020, −0.019]	−0.018 [−0.019, −0.017]

mean harm drifts from 0.458 to 0.421 across the sweep, which is why each row is read against its own mean. The mean drifts although the variance of the harm logit is fixed, because z_{spend} is right-skewed and the mean of a sigmoid of a skewed logit depends on its shape as well as its variance.

The gap depends on the budget (Table 3). At $\kappa = 0.10$ the estimated rule's gap is +0.152 [+0.146, +0.158], and the soft weight and the hard cap leave gaps of +0.018 [+0.015, +0.021] and +0.017 [+0.014, +0.021], above the population mean. Table D.2 repeats the sweep at $\kappa = 0.10$. Ranking on the estimated effect alone, which reads no harm measure, has a gap of −0.005 [−0.009, −0.002] at $\kappa = 0.30$, an exposure of 0.416, and −0.012 [−0.017, −0.008] at $\kappa = 0.10$. In this design it therefore reaches exposure comparable to the harm-aware rules with no harm input, which is a property of a design in which the effect falls with spend, not a safeguard.

That the sign follows the harm-value association is a finding about the main design,

Table 3: Harm exposure of the treated minus the population mean by budget and harm design ($c_{\text{harm}} = 1$, $\sigma_h = 0$, 95% confidence intervals in brackets). The population mean is that of the row's design. Panel A uses estimated components and panel B feasible-oracle components. Harm-blind, soft weight and hard cap rank on $\hat{\tau}\hat{V}$ or on the feasible-oracle product; CATE only ranks on the effect alone and reads no harm measure.

Harm design	Rule	$\kappa = 0.10$	$\kappa = 0.30$
Panel A: estimated components			
coupled	Harm-blind (τV)	+0.152 [0.146, 0.158]	+0.055 [0.051, 0.058]
coupled	Soft weight	+0.018 [0.015, 0.021]	−0.008 [−0.010, −0.005]
coupled	Hard cap	+0.017 [0.014, 0.021]	−0.008 [−0.010, −0.005]
coupled	CATE only	−0.012 [−0.017, −0.008]	−0.005 [−0.009, −0.002]
decoupled	Harm-blind (τV)	+0.142 [0.136, 0.149]	+0.039 [0.036, 0.042]
decoupled	Soft weight	+0.007 [0.003, 0.010]	−0.021 [−0.024, −0.019]
decoupled	Hard cap	+0.006 [0.002, 0.009]	−0.022 [−0.024, −0.019]
decoupled	CATE only	−0.022 [−0.027, −0.017]	−0.018 [−0.022, −0.015]
Panel B: feasible-oracle components			
coupled	Harm-blind (τV)	+0.052 [0.051, 0.054]	+0.033 [0.032, 0.034]
coupled	Soft weight	+0.001 [0.000, 0.002]	−0.012 [−0.013, −0.011]
coupled	Hard cap	+0.000 [−0.001, 0.001]	−0.012 [−0.013, −0.011]
coupled	CATE only	−0.068 [−0.069, −0.067]	−0.046 [−0.047, −0.045]
decoupled	Harm-blind (τV)	−0.018 [−0.019, −0.017]	−0.014 [−0.015, −0.013]
decoupled	Soft weight	−0.047 [−0.048, −0.046]	−0.046 [−0.046, −0.045]
decoupled	Hard cap	−0.047 [−0.048, −0.046]	−0.046 [−0.046, −0.045]
decoupled	CATE only	−0.125 [−0.126, −0.124]	−0.087 [−0.087, −0.086]

not a general rule, and Table 3 holds two counterexamples at $c_{\text{harm}} = 1$. First, with feasible-oracle components effect-only ranking has a gap of −0.046 [−0.047, −0.045]. The effect falls with spend, which drives harm, so ranking on the effect steers away from harm; for a rule that ranks on the effect alone only the second term of the decomposition in the proof of Proposition 2 remains, and condition (B) fails. Second, in the decoupled design of Section 6.4 the feasible-oracle τV rule has a gap of −0.014 [−0.015, −0.013], below the population mean, although the correlation of harm and value is +0.82. There harm depends on the covariates only through structural value, so condition (A) holds up to the clip on value, but the effect still falls with spend, (B) fails, and its negative term outweighs the positive one. The estimated rule in the same design has a gap of +0.039 [+0.036, +0.042], so whether value-weighted ranking concentrates harm depends on how the effect and its estimate co-move with harm, not on the harm-value association alone. We did not compute the two terms of the decomposition for each rule, so this reading of the counterexamples is an inference from the design, not a measurement.

6.3. Harm-aware rules and the cost of safety

With estimated components at $c_{\text{harm}} = 1$ and $\sigma_h = 0$, the soft harm weight cuts exposure to 0.414, below the population mean of 0.421, and raises operator welfare to 19,045 from the harm-blind 16,446 (Figure 1). The paired difference is +2,598 [2,406, 2,790], $d = 1.89$. The hard cap reaches the same measured exposure, 0.413, at welfare 15,701. Hard cap minus soft weight is −3,344 [−3,539, −3,148], $d = -2.38$, so with estimated components the cap is the more expensive rule. Relative to harm-blind welfare,

the soft weight raises welfare by 15.8% and the cap costs 4.5%. The soft weight also beats CATE-only ranking, by +2,411 [2,239, 2,583], $d = 1.95$, or +14.5%. The gain is therefore not shrinkage toward effect-only ranking, which Section 6.6 finds has the higher mean welfare for every learner at these budgets. At $\kappa = 0.10$ the soft weight earns 5,968, against 3,284 for the harm-blind rule, 3,894 for the hard cap and 4,766 for CATE-only ranking.

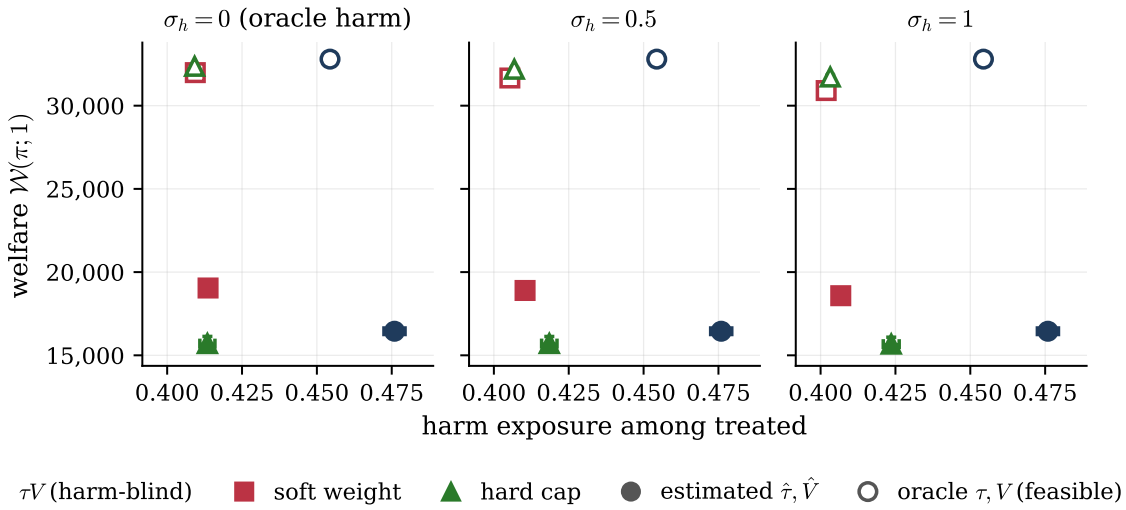


Figure 1: Operator welfare against harm exposure among the treated ($\kappa = 0.30$, $c_{\text{harm}} = 1$), one panel per harm-measure noise σ_h. Filled markers use estimated components and open markers the feasible oracle.

The order reverses with feasible-oracle components. There the cap earns 32,361 and the soft weight 31,986, against 32,797 for the harm-blind rule; hard cap minus soft weight is +376 [326, 426], $d = 1.05$. The cap's safety cost is +1.3% against +2.5% for the soft weight. The cap is therefore cheaper only with feasible-oracle components, the setting closest to the population statement of Corollary 2. Appendix D gives welfare for every cell of the sweep.

The ceiling holds only for the harm measure the rule is fed. When $\hat{H}$ is noisy, true exposure exceeds measured exposure for both harm-aware rules. At $\sigma_h = 0.5$ the overshoot is +0.013 [+0.012, +0.014] for the hard cap and +0.005 [+0.004, +0.005] for the soft weight, and at $\sigma_h = 1$ it is +0.043 [+0.041, +0.044] and +0.026 [+0.024, +0.027]. The hard cap exceeds its ceiling on true harm in 100.0% of replications at $\sigma_h = 0.5$ and 100.0% at $\sigma_h = 1$, but the binary rate says less than the size of the overshoot. The mechanism is selection: both harm-aware rules favour players with low $\hat{H}$, which over-represents negative measurement error, and the noise on the logit scale also pulls $\hat{H}$ toward 0.5. With estimated components the soft weight overshoots less than the cap at both noise levels, and it bounds nothing by design (Section 5).

The welfare gain is not a harm effect. We do not attribute the soft weight's welfare gain with estimated components to harm information. Three results in the sweep point to a different mechanism. First, the soft weight still beats harm-blind ranking where harm falls steeply with value, although by far less: at $c_{\text{harm}} = -1$ the difference is +468 [419, 517], Holm $p < 0.001$. Where harm falls weakly with value there is no gain: at $c_{\text{harm}} = -0.5$

the difference is −66 [−125, −6], Holm $p = 0.0638$. Second, where harm is independent of value the gain disappears at $\kappa = 0.30$: at $c_{\text{harm}} = 0$ it is −60 [−135, 16], Holm $p = 0.120$. At $\kappa = 0.10$ a smaller gain remains, +312 [255, 369], Holm $p < 0.001$, although harm is then independent of every covariate and so of the effect; demoting players at random helps when the top of the $\hat{\tau}\hat{V}$ ranking is poor, which it is at that budget. Third, with feasible-oracle components, where the product score is accurate, the soft weight costs +2.5% and the additive cap, which does not scale the product, costs +1.3%. The gain is therefore large only when the harm measure rises with value and the product score is inaccurate, small when harm falls steeply with value, and absent at $\kappa = 0.30$ when harm falls weakly with value or is independent of it, which is the pattern a regulariser on the upper tail of $\hat{\tau}\hat{V}$ would produce. In this design the effect falls with spend (Section 6.1), so the players with the largest values tend to have small or negative effects. A noisy $\hat{\tau}$ multiplied by a large $\hat{V}$ promotes exactly those players, and a weight that falls with a value-tracking harm measure trims them; at $c_{\text{harm}} = -1$ the same weight trims low-value players instead, and the gain is smaller. We did not run placebo rules that temper value without any harm information, such as a value cap, so the pattern supports this reading without isolating it. The practical consequence is that the soft weight should be chosen for what it does to harm exposure, and its welfare gain should not be expected where the effect does not fall in the value tail.

6.4. A decoupled-harm test

The soft weight's gain over harm-blind ranking could also come from the harm design. In the main design harm and the effect both load on spend, so $\hat{H}$ may carry information about τ beyond value that the harm weight exploits. The decoupled design removes that direct channel. Its harm logit is $-0.3 + \sigma_L\,(c_{\text{harm}}\,\zeta_i + \sqrt{1 - c_{\text{harm}}^2}\,Z_i) + e_i$, where ζ_i is the standardised noise-free structural value $V_i^\circ = 10\,\text{spend}_i + 40\,\text{login}_i + 5\,\text{age}_i$, so harm depends on X_i only through V_i°. The structural value V° differs from $\bar{V}(x) = \mathbb{E}[V_i \mid X_i = x]$ of Remark 5 only through the clip of V to $[50, 8{,}000]$. The population, effect, value, treatment and outcomes are those of the main design, and the logit variance and the draws of Z_i, e_i and the measurement noise are unchanged. At $c_{\text{harm}} = 1$ the partial correlation of τ and the harm logit given V° is +0.000 in the decoupled design against +0.245 in the main design, and given the realised V it is −0.008 against +0.222. The correlation of harm and realised value is similar in the two designs (+0.82 and +0.80).

Table 4: The soft weight against harm-blind and CATE-only ranking under coupled (main) and decoupled harm ($\kappa = 0.30$, $c_{\text{harm}} = 1$, estimated components, 95% confidence intervals in brackets). Welfare differences are paired over replications. The exposure gap is the soft weight's treated mean true harm minus the population mean harm of that design.

Harm design	σ_h	Soft minus harm-blind	Soft minus CATE only	Exposure gap
coupled	0.0	+2,598 [2,406, 2,790]	+2,411 [2,239, 2,583]	−0.008 [−0.010, −0.005]
decoupled	0.0	+3,246 [3,040, 3,452]	+3,059 [2,878, 3,240]	−0.021 [−0.024, −0.019]
coupled	0.5	+2,450 [2,258, 2,642]	+2,263 [2,092, 2,434]	−0.011 [−0.013, −0.008]
decoupled	0.5	+3,147 [2,942, 3,351]	+2,959 [2,782, 3,137]	−0.024 [−0.027, −0.022]
coupled	1.0	+2,135 [1,940, 2,331]	+1,948 [1,774, 2,123]	−0.015 [−0.017, −0.012]
decoupled	1.0	+2,922 [2,717, 3,127]	+2,735 [2,554, 2,917]	−0.029 [−0.031, −0.026]

Table 4 shows that the gain persists and grows. At $\sigma_h = 0$ the soft weight beats

harm-blind ranking by +2,598 in the main design and by +3,246 [3,040, 3,452] in the decoupled design, $d = 2.20$. Against CATE-only ranking the difference is +2,411 in the main design and +3,059 [2,878, 3,240] in the decoupled design, or +14.5% and +18.4% [+17.0%, +19.7%] of CATE-only welfare, with Holm-adjusted $p < 0.001$ within the cell's eight paired tests. The same holds at $\sigma_h = 0.5$ and 1, and the soft weight's exposure gap is more negative in the decoupled design. With feasible-oracle components the soft weight still beats CATE-only ranking in both designs (+1,138 and +1,482), but it trails harm-blind ranking, by −811 and −468, so decoupling does not remove the cost of safety with accurate components.

The test establishes a limited result. In the decoupled design harm carries no information about τ beyond V°, so the gain that persists works through value. That fits the regularisation reading of Section 6.3: $\hat{H}$ is still a monotone function of structural value plus noise, so $\omega(\hat{H})$ still trims the upper value tail, where the effect is small or negative because it falls with spend. A complementary route, which we did not test, is that $\hat{H}$ proxies the structural value that the misspecified $\hat{V}$ misses, since the proxy model drops age and flattens spend. The test therefore rules out a direct link between harm and the effect beyond value as the source of the gain, and neither remaining route makes the gain a consequence of measuring harm.

6.5. Operator value, benchmarks and the budget diagnostic

Table 5 reports operator welfare at $\kappa = 0.30$. The churn-weighted rules of the secondary comparison show that the default preference weight costs value: Churn-weighted (est.) earns 13,646 against 16,446 for CATE × LTV, a difference of −2,801 [−2,923, −2,678], or −17.0%. The calibrated rule, Churn-weighted (cal., AIPW), recovers part of that gap (15,566) but stays below the identity. AIPW calibration picks the identity in 13 of 200 replications (6.5%) and costs −881 [−1,217, −544], $d = -0.37$, against the identity. The plug-in calibrated row equals CATE × LTV exactly, because the plug-in objective selects the identity in 200 of 200 replications by Observation 1 (Appendix C).

Table 5: Operator welfare by strategy (10,000 players, 200 replications, $\kappa = 0.30$). Welfare is operator welfare $\mathcal{W}(\pi; \mathbf{1})$ on the test set, SD is the standard deviation across replications, and risk exposure is mean churn probability among the treated. Churn-weighted rows, a secondary comparison of the preference weight g, use θ_0 or the calibrated θ as named, and the S-learner uses the proxy value model.

Strategy	Welfare	SD	Risk exposure
Random	5,399	1,362	0.166
CATE only	16,634	2,962	0.153
CATE × LTV	16,446	3,115	0.136
Churn-weighted (est.)	13,646	3,026	0.146
Churn-weighted (cal., AIPW)	15,566	4,029	0.142
Churn-weighted (cal., plug-in)	16,446	3,115	0.136
S-learner	24,273	4,163	0.120
Oracle, hindsight (g=1)	44,615	807	0.138
Oracle, feasible (g=1)	32,797	791	0.123
Oracle, hindsight (θ_0)	36,641	705	0.192
Oracle, feasible (θ_0)	28,178	705	0.175

The feasible oracle reaches only 73.5% [73.3%, 73.7%] of the hindsight oracle, so about

a quarter of hindsight welfare is unreachable from covariates, and we measure estimation cost against the feasible oracle. Against it, the AIPW-calibrated rule, Churn-weighted (cal., AIPW), reaches 47.5% [45.8%, 49.2%] in level and 37.0% after subtracting Random and rescaling. CATE × LTV reaches 50.1% [48.8%, 51.5%] (40.2% normalised) and the S-learner 74.0% [72.3%, 75.8%].

Table 6 varies the budget. CATE × LTV minus CATE only is −1,482 [−1,632, −1,332] at $\kappa = 0.10$, −842 [−998, −687] at 0.20, −187 [−319, −55] at 0.30, and +195 [133, 256] at 0.50. Value weighting hurts at tight budgets and helps at generous ones. With feasible-oracle effects and values it helps at every budget: at $\kappa = 0.10$ the difference is +1,441 [1,383, 1,499], $d = 3.46$.

Table 6: Operator welfare by budget. CATE only and CATE × LTV use the default T-learner with the proxy value model; Cal. is Churn-weighted (cal., AIPW); the oracle is the feasible oracle with $g \equiv 1$. Holm p-values test Cal. against the named strategy.

	Operator welfare				Holm p, Cal. vs	
κ	CATE only	CATE × LTV	Cal. (AIPW)	Feasible oracle	CATE only	CATE × LTV
0.10	4,766	3,284	3,155	13,233	<0.001	0.143
0.20	10,750	9,908	9,363	23,864	<0.001	<0.001
0.30	16,634	16,446	15,566	32,797	<0.001	<0.001
0.50	26,740	26,935	24,987	46,046	<0.001	<0.001

The crossover budget $\kappa^\star$ is where this penalty ends (Table 7), and it belongs to the learner. Crossing every effect source with every value source isolates its origin. With feasible-oracle effects and values there is no crossing: value weighting leads at every grid budget, and $\kappa^\star$ is < 0.050 with a crossing in 0.0% of draws. With the T-learner and the proxy value model $\kappa^\star$ is 0.350 [0.325, 0.375], with a crossing in 100.0% of draws. With the X-learner and the proxy it is 0.425 [0.325, > 0.500], crossing in 66.9% of draws, and with the S-learner it never crosses within the grid (> 0.500). For the T-learner with feasible-oracle or well-specified values the point estimate sits above the grid and only 43.2% and 24.9% of draws cross inside it. Our explanation is a hypothesis we did not test: multiplying a noisy effect estimate by a value estimate with large spread lets the value estimate dominate the ranking in the tail where the effect is least well measured, and a smoother value estimate or a less noisy effect estimate weakens that. Part of the penalty may also come from the falling effect in the value tail described in Section 6.3, and we did not separate the two.

6.6. Learner choice

Table 8 compares the T-, X-, S- and DR-learners, default and tuned, under both value models and with no value model. Effect-only ranking beats $\hat{\tau}\hat{V}$ in welfare for every learner, tuning and value model at every budget up to $\kappa = 0.30$. The difference is Holm-significant in all but one of these cells, the default DR-learner with the proxy value model at 0.30, where it is +102 [−33, 237]. For the default T-learner with the proxy value model the gain from dropping $\hat{V}$ is +1,482, +842 and +187 at $\kappa = 0.10$, 0.20 and 0.30, and for the tuned S-learner it is +3,372 [2,815, 3,928] at 0.30, $d = 0.84$. These tests form their own Holm family of sixteen comparisons per budget, separate from the learner comparisons in

Table 7: Crossover budget $\kappa^\star$ by effect source and value source. $\kappa^\star$ is the smallest budget on a 0.025 grid from which $\hat{\tau}\hat{V}$ beats $\hat{\tau}$ alone at every larger grid budget; < and > mark values censored at the grid ends. Below, Crossing and Above count the 2,000 bootstrap draws whose $\kappa^\star$ lies below the grid (value weighting leads throughout), inside it, or above it (effect-only ranking leads throughout). Share is the proportion of draws that cross, and Sign changes counts sign changes of the point difference curve.

		Bootstrap draws						
τ source	V source	Below	Crossing	Above	$\kappa^\star$	95% CI	Share	Sign changes
feasible oracle	feasible oracle	2,000	0	0	<0.050	[<0.050, <0.050]	0.0%	0
feasible oracle	proxy	2,000	0	0	<0.050	[<0.050, <0.050]	0.0%	0
feasible oracle	well-specified	2,000	0	0	<0.050	[<0.050, <0.050]	0.0%	0
T-learner	feasible oracle	0	865	1,135	>0.500	[0.450, >0.500]	43.2%	0
T-learner	proxy	0	2,000	0	0.350	[0.325, 0.375]	100.0%	1
T-learner	well-specified	0	497	1,503	>0.500	[0.475, >0.500]	24.9%	0
X-learner	feasible oracle	0	0	2,000	>0.500	[>0.500, >0.500]	0.0%	0
X-learner	proxy	0	1,338	662	0.425	[0.325, >0.500]	66.9%	1
X-learner	well-specified	0	0	2,000	>0.500	[>0.500, >0.500]	0.0%	0
S-learner	feasible oracle	0	0	2,000	>0.500	[>0.500, >0.500]	0.0%	0
S-learner	proxy	0	0	2,000	>0.500	[>0.500, >0.500]	0.0%	0
S-learner	well-specified	0	0	2,000	>0.500	[>0.500, >0.500]	0.0%	0

the table. At $\kappa = 0.50$ the default T-, X- and DR-learners with the proxy value model reverse sign (-195 [-256, -133], -20 and -79), and only the T-learner's reversal is Holm-significant. Even at $\kappa = 0.50$, effect-only ranking has the higher mean welfare in 13 of 16 learner cells, 12 of them Holm-significant. In the main design, then, value weighting pays only when its components are accurate: it wins with feasible-oracle components, while with estimated effects it pays only at a generous budget and only for some learners, and never for the S-learner on the grid. This is a property of the design, not of the rule. Its boundary is the null scenario of Table E.1: without the effect-churn moderation, which shrinks the effect for low-churn players, value weighting wins with the default T-learner by +5,157 [4,704, 5,610]. It also wins in the misspecified-effect (+287) and asymmetric-moderation (+398) scenarios. Which regime an operator is in depends on effect size, training sample size and value spread, which we did not vary jointly (Section 8).

Under R-loss tuning the T-learner chooses a heavily shrunk depth-1 fit; its estimated heterogeneity is strongly attenuated. The tuned T- and DR-learner fits sit at the grid edge toward more shrinkage, and the S-learner at the edge toward less (Appendix E); the evidence is consistent with a weak-signal regime rather than under-tuning, but a wider grid could change the magnitudes. With tuned learners and the proxy value model, the S-learner earns +5,496 [4,651, 6,341] more than the T-learner at $\kappa = 0.30$, $d = 0.91$, or +32.4% [+26.2%, +38.6%]. The DR-learner equals the R-learner here because with a known constant propensity 0.5 the pseudo-outcomes coincide and the R weights are constant. Three caveats apply to tuning. The spread of the chosen R-loss across learners is only 0.00005, tuning used only 5 pilot replications, and R-loss rewards calibration of the effect estimate, not the ranking that targeting needs. The highest welfare of any estimated rule comes from the S-learner, one model trained on Y with treatment as a feature, ranked on the effect alone without a value model (25,809 tuned, 25,351 default); we have not tested it against the other learner and rule cells, so this is descriptive.

The value model matters in a way that contradicts a natural expectation. The proxy value model beats the well-specified one: CATE × LTV with the well-specified model

Table 8: Learner comparison at $\kappa = 0.10$ and 0.30 (200 replications). Welfare is operator welfare, and value model "none" ranks on the estimated effect alone. Holm p tests each cell against the default T-learner with the same value model. Shapley is the order-free share of regret against the feasible oracle attributed to the effect estimate, with a 95% confidence interval. The DR-learner equals the R-learner under the known propensity 0.5.

		$\kappa = 0.10$		$\kappa = 0.30$		
Learner	Value model	Welfare	Holm p	Welfare	Holm p	Shapley CATE share
T tuned	proxy	3,125	1.000	16,941	0.233	100.6% [100.2%, 101.1%]
T tuned	well-specified	2,950	1.000	16,605	0.216	100.1% [100.0%, 100.2%]
T tuned	none (CATE only)	6,211	<0.001	18,760	<0.001	–
T default	proxy	3,284	–	16,446	–	101.1% [100.9%, 101.3%]
T default	well-specified	3,033	–	15,914	–	99.9% [99.8%, 100.0%]
T default	none (CATE only)	4,766	–	16,634	–	–
X tuned	proxy	6,869	<0.001	22,741	<0.001	101.8% [101.3%, 102.3%]
X tuned	well-specified	6,541	<0.001	22,243	<0.001	100.1% [99.9%, 100.3%]
X tuned	none (CATE only)	8,854	<0.001	24,218	<0.001	–
X default	proxy	5,386	<0.001	20,212	<0.001	101.2% [101.0%, 101.5%]
X default	well-specified	5,035	<0.001	19,729	<0.001	99.9% [99.8%, 100.0%]
X default	none (CATE only)	6,612	<0.001	20,456	<0.001	–
S tuned	proxy	5,949	<0.001	22,437	<0.001	102.0% [101.5%, 102.5%]
S tuned	well-specified	5,606	<0.001	21,929	<0.001	99.9% [99.7%, 100.1%]
S tuned	none (CATE only)	8,768	<0.001	25,809	<0.001	–
S default	proxy	7,585	<0.001	24,273	<0.001	100.6% [100.2%, 101.1%]
S default	well-specified	7,294	<0.001	24,015	<0.001	99.9% [99.8%, 100.1%]
S default	none (CATE only)	8,874	<0.001	25,351	<0.001	–
DR tuned	proxy	3,004	0.954	17,639	0.0203	101.9% [101.4%, 102.4%]
DR tuned	well-specified	2,712	0.887	16,956	0.0668	100.1% [100.0%, 100.2%]
DR tuned	none (CATE only)	7,352	<0.001	21,693	<0.001	–
DR default	proxy	3,791	<0.001	16,758	0.0879	101.1% [100.9%, 101.2%]
DR default	well-specified	3,454	<0.001	16,239	0.0668	99.9% [99.8%, 100.0%]
DR default	none (CATE only)	4,715	1.000	16,861	0.216	–

earns −532 [−573, −491] relative to the proxy, $d = -1.81$. A hypothesis consistent with the variance-amplification argument above is that the proxy, which drops age and flattens spend, gives a smoother value estimate and so amplifies less of the effect estimate's noise. It is an instance of the point of Fernández-Loría and Provost (2022) that the quality of a targeting decision need not track the accuracy of the estimates behind it. The Shapley split attributes essentially all regret to the effect estimate, 99.9% with the well-specified value model and 101.1% with the proxy. A share above 100% means that, averaged over orders, swapping in $\hat{V}$ slightly offsets error in $\hat{\tau}$. One reading, consistent with Section 6.3, is that the flatter proxy $\hat{V}$ helps in the upper value tail, where the effect is small or negative.

7. Responsible deployment

Sections 5 and 6 describe what the rules do to harm in a simulation. This section sets out what an operator needs in place before any of them is used on a live campaign. The measures below are preconditions, because a value-weighted score does not detect harm on its own.

The harm proxy. The harm measure $\hat{H}$ in this paper is a score built from behavioural markers in account data, such as deposit velocity, escalating session length and loss chasing (Braverman and Shaffer, 2012; Auer and Griffiths, 2023; Delfabbro et al., 2025). We use behavioural markers because harm is not confined to players who screen positive on a clinical instrument. It is spread across risk levels (Browne et al., 2016), and its measurement is contested (Delfabbro and King, 2019). The Problem Gambling Severity Index is a survey instrument that few players in an operator's base have completed, so it can validate a marker score (Delfabbro et al., 2025) but cannot serve as the operator's exclusion list. The proxy is a population-level input to targeting, to be audited against harm outcomes the operator can confirm, and not a diagnosis of any individual.

Order of operations. Players who have self-excluded, or whom the operator's customer-interaction process has flagged for harm indicators, form an exclusion set $\mathcal{E}$ that is removed before any score is computed. The rule then ranks the eligible players only, $\mathcal{S}^* = \arg\,\text{top-k}_{i \notin \mathcal{E}}\, s_i$. The harm weight of Corollary 1 sits inside the objective, so that harm trades against value for every player who remains. Filtering after ranking is no substitute, because the ranking has already spent the budget. Two defaults follow. The system fails closed: when harm signals are missing for part of the population, the high-value tail of that part is excluded and not scored harm-blind. Exclusion and harm signals are re-derived at every campaign, because a player outside $\mathcal{E}$ at one round may have crossed a threshold by the next. For the preference weight of Appendix B we recommend the floor $g_{\min} = 0$ in live use, so that the highest-churn players receive no extra weight. The simulation uses $g_{\min} = 0.20$ for contrast, and any positive floor needs explicit justification, because the highest-churn players are the most likely to be disengaging for reasons connected with harm.

Soft weight or hard cap. With estimated components we prefer the soft harm weight, and we recommend it for harm reduction only. It lowers exposure at both budgets we test, although at $\kappa = 0.10$ not to the population mean (Section 6.2). Its welfare gain in our simulation is not attributable to harm information: it comes from trimming the noisy upper tail of the product score in a design where the effect falls in that tail (Section 6.3), so an operator should not count on it, and with accurate components the soft weight costs value. A hard cap earns more only when its components are accurate, and its ceiling holds only for the harm measure it is fed (Remark 7). Where a guarantee is required, use the exclusion set, which is the limit of the cap at a ceiling of zero, or a cap on a harm measure that has been audited. In our main design, ranking on the effect alone reached similar exposure with no harm input; that is a feature of the design, not a safeguard an operator can rely on.

Harm-model error. The harm measure is itself an estimate, and Section 6.3 shows what noise in it does: under both harm-aware rules true exposure exceeds measured exposure, the overshoot grows with the noise, and the hard cap breaches its ceiling on true harm. An operator should therefore set the operational ceiling below the target by the expected overshoot, and estimate that overshoot on a holdout by comparing the marker score with confirmed harm outcomes. These sizes belong to our simulated noise model, and an operator's own overshoot has to be measured.

Evaluation and monitoring. Operator value and harm should be re-estimated each campaign with a doubly robust (AIPW) estimator (Dudík et al., 2011) on a randomised

holdout, which keeps the propensities known; counterfactual risk minimisation (Swaminathan and Joachims, 2015) is an alternative. The treatment-effect estimates used in the score can serve as the outcome model, so targeting and evaluation share components. Players in $\mathcal{E}$ are excluded from the holdout as well as from the ranking, because randomising an at-risk player into treatment is what the safeguards forbid. The holdout therefore supports estimates only for the eligible population, and its membership is re-derived each round together with $\mathcal{E}$.

Regulatory mapping. Table 9 sets the framework against the requirements most relevant in Great Britain. The Gambling Commission announced its new rules on promotions on 26 March 2025, and they took effect on 19 January 2026 (Gambling Commission, 2025). Inducements are also a recognised harm-minimisation concern in the research literature (Hing et al., 2017; Newall et al., 2025). The table records what the framework addresses and is not a compliance assessment. The methods can also be misused to direct incentives at harmed players, which is why the dual-use statement below ties publication to these safeguards.

8. Discussion

8.1. Why estimated value weighting does not pay here

The oracle result and the estimated result point in opposite directions, and the difference is estimation. Lemma 1 locates regret in players misranked near the threshold; heuristically, in a product score the error in $\hat{\tau}$ is scaled by $\hat{V}$, and V varies by orders of magnitude, so noise in $\hat{\tau}$ is amplified before it ranks players. Effect-only ranking avoids that amplification, and the Shapley split puts nearly all regret on $\hat{\tau}$ (Section 6.6). The value model contributes in an unexpected direction: the proxy beats the well-specified model (Section 6.6), which fits this amplification reading if the smoother proxy passes on less of the effect estimate's noise. Variance is not the only candidate mechanism: in this design the effect falls in the upper value tail, which a product score with a large $\hat{V}$ over-weights, and the same feature explains the soft weight's welfare gain (Section 6.3). The tight-budget penalty of value weighting, and the budget at which it starts to pay, therefore belong to the learner and the design, not to the rule, and the null scenario, where value weighting wins, marks the edge of the design in which we observe them.

8.2. Calibrating the preference weight

No preference weight we calibrate beats the identity. Plug-in calibration selects the identity by construction (Observation 1), so its selection rate is a consistency check and not a finding. AIPW calibration, which nothing forces toward the identity, rarely selects it and costs operator value against it (Section 6.5). The operator sets g from preferences, and calibration prices the choice (Appendix B and Appendix C).

8.3. Scope

Every optimality result is conditional on operator value being multiplicatively separable in effect, value and preference weight (Definition 1). That is the form the $\hat{\tau}\hat{V}$ heuristic already takes, and it is what permits a decomposition into separately auditable components. It fails if, for example, an operator's willingness to take risk exposure varies with customer value, so that the preference factor is $g(R, V)$ and the score no longer factors; such an operator needs a more general objective. The binary outcome restricts the framework to the extensive margin. A continuous outcome changes the notation and not the

Table 9: Regulatory requirements in Great Britain and how the framework addresses them.

Requirement	Source	How the framework addresses it
Stop marketing and the uptake of new bonus offers once strong indicators of harm are identified	Customer interaction guidance for remote licensees, Requirement 10 (Gambling Commission, 2022); in effect from 31 October 2023	Addressed in part. The exclusion set and the fail-closed default stop incentives to flagged players before ranking. Flagging depends on the operator's indicators, which the framework does not supply.
Wagering requirements capped at 10 times; one product type per incentive; terms clear, transparent and fair	SR code 5.1.1, Rewards and bonuses (Gambling Commission, n.d.), as revised under the rules announced 26 March 2025 and in force 19 January 2026 (Gambling Commission, 2025)	Not addressed. The framework decides who receives an incentive and not its terms. Compliant terms are a precondition outside the model, and operator value in the simulation ignores wagering terms. An operator must check its offer design against the final rule text.
High-value or VIP incentives consistent with the licensing objectives	SR code 5.1.1, Rewards and bonuses (Gambling Commission, n.d.)	Addressed in part. Value weighting is the practice this requirement bears on. The diagnostic of Proposition 2 tests whether a rule concentrates incentives on harmed players, and the harm weight reduces the concentration. Neither establishes consistency with the licensing objectives.
Harm-focused reform of online gambling, including affordability checks	High Stakes white paper, CP 835, April 2023 (Department for Digital, Culture, Media and Sport, 2023)	Not addressed. The framework uses no affordability data. Affordability signals can enter the harm measure where an operator holds them.

results, and V may then be redundant. The auxiliary components must be pre-treatment (Remark 1), which sequential deployment weakens because later-period value and churn reflect earlier treatments. Costs are homogeneous here, but the realised cost of a bonus depends on the player's response, and Haupt and Lessmann (2022) derive a profit rule for that case, which we do not simulate. SUTVA rules out spillovers between players who interact.

Harm is treated as pre-treatment, so the incentive's own effect on harm is not modelled. A retention incentive that raises play among harmed players would add to the concentration shown here. The evidence is a simulation with illustrative parameters and no operator data. The public uplift benchmarks cannot test the central claims, because they carry neither a per-player value nor a harm measure, and genuine validation needs operator data with both.

8.4. Limitations

The harm model is stylised. True harm is a continuous score on $[0, 1]$ with a population mean of 0.421 at $c_{\text{harm}} = 1$, not a construct calibrated to the literature, in which problem-gambling prevalence is a few percent and harm severity is skewed. Exposure gaps on this scale show direction and relative size, not a clinical quantity, and the soft weight's thresholds $[0.6, 0.9]$ are tied to this distribution and were not varied. The operator's harm measure is a noisy copy of true harm rather than a model estimated from observable markers, so every noise level we test is more favourable than a real marker score. The hard cap is evaluated at a single ceiling, matched to the soft weight's measured exposure, so the comparison of the two rules is one point on each rule's welfare-exposure frontier and not a comparison of frontiers. Harm enters only as a weight or a constraint and never in the objective, so we cannot say which welfare-harm trade-off either rule achieves.

The mechanism behind the soft weight's welfare gain is inferred, not isolated. We did not run placebo rules that temper value without harm information, rerun the harm study in designs where the effect does not fall with value, or rerun it with learners other than the default T-learner. Nor did we compute the two terms of the covariance decomposition of Proposition 2 for each rule, so our reading of the counterexamples in Section 6.2 as failures of condition (B) is not measured.

The comparison of value weighting with effect-only ranking comes from one design. Its sign reverses in the null scenario, and a factorial sweep over effect size, training sample size, value spread and the noise in the effect and value would map where each rule wins. Linking that map to a signal-to-noise quantity an operator can estimate, such as a rank-weighted average treatment effect (Yadlowsky et al., 2025), is future work.

The preference-weight study has a mismatch between the weight and its input. The churn probability p_i on which g is defined is generated independently of baseline retention b_i, while the churn model is trained on control-arm non-retention, which estimates $1 - b_i$ and not p_i, so the calibration results of Appendix C test a weight applied to a proxy that does not track its target.

Inference is over Monte Carlo replications of one design, so small p-values measure simulation precision rather than practical importance, and welfare is a sum over the test players in arbitrary value units. Top-κ selection breaks ties in sort order without an explicit rule, and the tuned learners produce coarse scores with many ties, which may add to their replication variance.

8.5. Extensions

Natural extensions are multi-armed and sequential treatment with bonus sizing, finite-sample regret bounds for the three-component product score in the manner of Athey and Wager (2021), and fairness criteria within the budget-constrained problem (Kitagawa and Tetenov, 2021; Viviano and Bradic, 2024).

9. Conclusion

Ranking retention incentives by estimated effect times value is not neutral about harm. For any rule that exhausts the budget, the harm exposure of the treated exceeds the population mean by the covariance between treatment and harm divided by the budget. Value-weighted ranking makes that covariance non-negative under sufficient conditions: harm rises with value among players with the same effect, and ranking on the effect does not steer away from harm. In our main design the gap for the estimated rule is +0.055

at $\kappa = 0.30$ and $+0.152$ at $\kappa = 0.10$, but harm rising with value is not enough on its own, and we report two cases in which a rule's gap is negative although harm rises with value. A harm weight on the score lowers exposure. Its welfare gain with estimated components comes from trimming the noisy value tail, where the effect is small or negative in this design, not from harm information, so the weight should be chosen for harm reduction and its welfare gain should not be expected to transfer. A hard cap is the population optimum for a ceiling, but it needs accurate inputs and guarantees only the harm that its measure sees, so with estimated components we prefer the soft weight and, where a guarantee is required, an audited harm measure and an exclusion set (Section 7).

In the main design, value weighting pays only when its components are accurate. With feasible-oracle effects and values it wins at every budget we test, and with estimated effects ranking on the effect alone yields more operator value for every learner at budgets up to $\kappa = 0.30$, significantly in all but one learner cell. The null scenario reverses this (Section 6.6), so the result belongs to the design, and mapping where each rule wins across effect size, sample size and value spread is future work. A feasible oracle that sees only covariates is the benchmark against which estimation cost should be read. All results come from a simulation with illustrative parameters and no operator data. Validating the diagnostic and the harm-aware rule needs operator data with observed lifetime value and a harm measure, and any deployment needs the safeguards of Section 7 whatever that validation shows.

CRediT authorship contribution statement

Mark Stent: Conceptualization, Methodology, Software, Formal analysis, Investigation, Visualization, Writing – original draft, Writing – review & editing.

Declaration of competing interest

The author is employed in the gambling and lottery industry. The framework studied here is not deployed at the author's employer, and there are no plans to deploy it there. The employer did not review this paper. No operator data informed any parameter of the data-generating process; all parameters are illustrative. The views expressed are the author's own.

Funding

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Ethics statement

This study involves no human participants, no personal data and no experiments on individuals. All results come from a synthetic, fully specified simulation (Section 6), and no real player or customer data was used, so no ethics-board approval was required. Because promotional targeting in online gambling can harm players, the paper reports that value-weighted targeting can concentrate incentives on harmed players (Section 6.2) and treats an exclusion set, a harm-aware score and a fail-closed default as preconditions for any deployment (Section 7).

Dual-use statement

The methods in this paper can be used to direct incentives more efficiently at players who are valuable to an operator, and Section 5 shows that such targeting can concentrate on harmed players. We publish the diagnostic and the harm-aware rule so that operators, regulators and auditors can detect and limit this effect. We do not support deploying value-weighted targeting without the safeguards of Section 7.

Data availability

This study uses no empirical data, and every reported quantity is generated by simulation. The code is in the public repository `https://github.com/markstent/value-weighted-targeting`. It holds the five simulation studies (main comparison, harm sweep, budget analysis, learner comparison and robustness scenarios), the table and macro generator that writes every number in this paper into generated files, the figure generator, and a run manifest that records the git commit, package versions and file hashes. The repository tracks the final run's manifest and result JSONs in `results/`, and `python src/run_all.py` reproduces them. Running `src/make_tex.py` and `src/make_figures.py` on the results regenerates every number, table and figure. The code is released under the MIT licence and the paper under CC BY 4.0. Each replication r uses the fixed seed $42 + 7r$, so the study is deterministic under the pinned dependencies.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the author used Claude (Anthropic) to review the manuscript, refactor and test the simulation code, check references, and edit text. After using this tool, the author reviewed and edited the content as needed and takes full responsibility for the content of the publication.

Appendix A. Proofs

Proof of Proposition 1

Proof. Write $F(c) = \mathbb{P}(s(X_i) > c)$, which is non-increasing and right-continuous, with $F(c) \to 0$ as $c \to \infty$. By (b), F is continuous on $(0, \infty)$. *Threshold.* If $F(0) = \kappa$, then $c^* = 0$ and $F(c^*) = \kappa$. If $F(0) > \kappa$, right-continuity gives $F(c) > \kappa$ for small $c > 0$, so $c^* > 0$; continuity of F at c^*, together with $F(c) > \kappa$ for $c < c^*$ and $F(c) \leq \kappa$ for $c > c^*$, gives $F(c^*) = \kappa$. Condition (a) excludes $F(0) < \kappa$, so in both cases $c^* \geq 0$ and $\mathbb{E}[\pi^*(X_i)] = \kappa$. *Optimality.* For any measurable $\pi : \mathcal{X} \to [0, 1]$ with $\mathbb{E}[\pi(X_i)] \leq \kappa$, adding and subtracting $c^* \, \mathbb{E}[\pi^* - \pi]$ gives

$$\mathcal{W}(\pi^*; g) - \mathcal{W}(\pi; g) = \mathbb{E}\Big[(\pi^*(X_i) - \pi(X_i))(s(X_i) - c^*)\Big] + c^*\Big(\kappa - \mathbb{E}[\pi(X_i)]\Big).$$

On $\{s > c^*\}$, $\pi^* - \pi = 1 - \pi \geq 0$; on $\{s < c^*\}$, $\pi^* - \pi = -\pi \leq 0$; on $\{s = c^*\}$ the factor $s - c^*$ is zero. The integrand is therefore non-negative, and the second term is non-negative because $c^* \geq 0$. *Uniqueness.* If π is also optimal, both terms vanish. The first gives $\pi = \pi^*$ almost surely on $\{s \neq c^*\}$. If $c^* > 0$, (b) gives $\mathbb{P}(s = c^*) = 0$ and we are done. If $c^* = 0$, then $\pi = 1$ almost surely on $\{s > 0\}$, so $\mathbb{E}[\pi] \geq F(0) = \kappa$; feasibility then forces $\pi = 0$ almost surely on $\{s = 0\}$, so again $\pi = \pi^*$ almost surely. □

Proof of Proposition 2

Proof. (i) Because $\pi(X_i)$ is a function of X_i, iterated expectations give $\mathbb{E}[\pi H] = \mathbb{E}[\pi \bar{H}]$ and $\mathbb{E}[H] = \mathbb{E}[\bar{H}]$. Hence $\eta(\pi) = \mathbb{E}[\pi \bar{H}]/\kappa$, and subtracting $\mathbb{E}[H] = \mathbb{E}[\pi]\,\mathbb{E}[\bar{H}]/\kappa$ gives (7). (ii) Write $A = a(X_i)$ and $B = b(X_i)$. By the law of total covariance,

$$\mathrm{Cov}(\pi, \bar{H}) = \mathbb{E}\Big[\mathrm{Cov}(\pi, \bar{H} \mid A)\Big] + \mathrm{Cov}\Big(\mathbb{E}[\pi \mid A],\ \mathbb{E}[\bar{H} \mid A]\Big).$$

The second term is non-negative by (B). For the first, fix $A = u$. If $u \leq 0$, then $uB \leq 0 \leq c$, so $\pi = 0$ and the conditional covariance is zero. If $u > 0$, then $\pi = \mathbf{1}\{B > c/u\}$ is a non-decreasing function of B, and since π is a function of (A, B), $\mathrm{Cov}(\pi, \bar{H} \mid A = u) = \mathrm{Cov}(\pi, \psi(u, B) \mid A = u)$. Two non-decreasing functions of the same random variable have non-negative covariance (Chebyshev's association inequality), so the first term is non-negative, and (i) gives $\eta(\pi) \geq \mathbb{E}[H_i]$. Under the strictness condition, the conditional covariance is strictly positive on a set of u of positive probability. □

Proof of Corollary 2

Proof. For any π satisfying both constraints,

$$\mathcal{W}(\pi^\ddagger) - \mathcal{W}(\pi) = \mathbb{E}\Big[(\pi^\ddagger - \pi)(s - \mu^* \hat{h} - \lambda^*)\Big] + \lambda^*\Big(\mathbb{E}[\pi^\ddagger] - \mathbb{E}[\pi]\Big) + \mu^*\Big(\mathbb{E}[\pi^\ddagger \hat{h}] - \mathbb{E}[\pi \hat{h}]\Big).$$

The first integrand is non-negative pointwise, as in Proposition 1. By complementary slackness the second term equals $\lambda^*(\kappa - \mathbb{E}[\pi]) \geq 0$ and the third equals $\mu^*(\eta_{\max}\kappa - \mathbb{E}[\pi \hat{h}]) \geq 0$. If π is also optimal, the first term vanishes, so $\pi = \pi^\ddagger$ almost surely off the boundary set $\{s - \mu^* \hat{h} = \lambda^*\}$, which has probability zero. □

Appendix B. Risk-modulation function, example and frontier

This appendix gives the piecewise linear preference weight g used in the simulation, a numerical example of the three scoring rules, and two remarks on how g trades operator welfare against the targeting pattern. The functional form is an implementation choice; the parameters are chosen for their operational reading.

The preference weight

For retention bonuses, the effectiveness of a bonus depends on the player's churn risk p (we write p for R here). Three regimes motivate the shape. Below p_L the player is stable and would likely stay without a bonus, so the weight is zero. Between the plateau bounds the player is at risk but reachable, so the weight is one. Above p_H the player has probably disengaged beyond what a bonus can reverse (Ascarza, 2018), so the weight falls to a floor $g_{\min}$ that reflects uncertainty in the churn estimate. The parameters $\theta = (p_L, p_{ML}, p_{MH}, p_H, g_{\min})$ have these direct meanings, and Figure B.1 plots g at the default $\theta_0 = (0.05, 0.40, 0.60, 0.90, 0.20)$ used in the simulation. The defaults are chosen for clear contrasts across regimes, and an operator should set θ from domain evidence. The weight g is a covariate-dependent targeting weight on an observable churn estimate, not risk aversion in the decision-theoretic sense of Manski and Tetenov (2007).

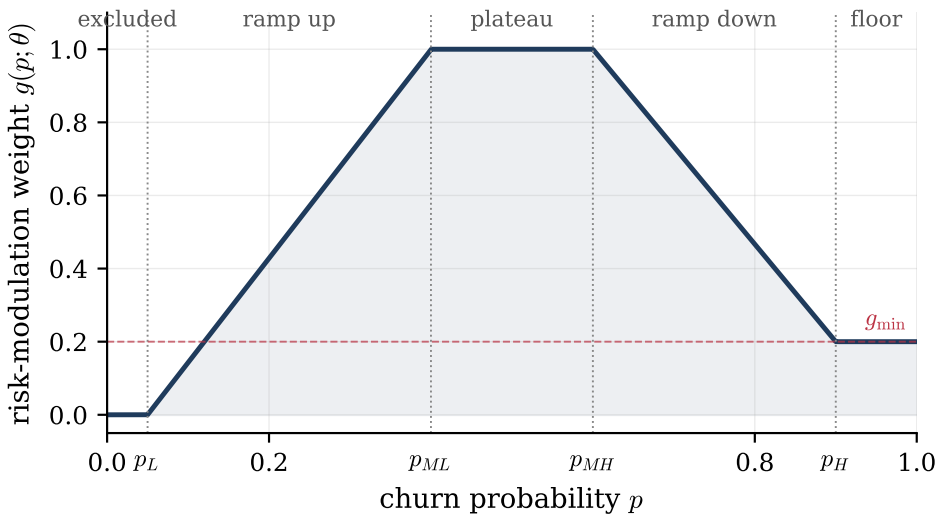


Figure B.1: The piecewise linear preference weight at default parameters $\theta_0 = (0.05, 0.40, 0.60, 0.90, 0.20)$. Players below p_L are excluded. The weight ramps to one through the moderate-risk plateau and falls to the floor $g_{\min}$ for the highest-churn players.

Definition 2 (Piecewise linear risk-modulation function)**.** The risk-modulation function $g : [0,1] \to [0,1]$ is defined as

$$g(p;\,\theta) = \begin{cases} 0 & \text{if } p < p_L, \\ \dfrac{p - p_L}{p_{ML} - p_L} & \text{if } p_L \le p < p_{ML}, \\ 1 & \text{if } p_{ML} \le p \le p_{MH}, \\ 1 - (1 - g_{\min})\,\dfrac{p - p_{MH}}{p_H - p_{MH}} & \text{if } p_{MH} < p < p_H, \\ g_{\min} & \text{if } p \ge p_H, \end{cases} \tag{B.1}$$

parameterised by $\theta = (p_L, p_{ML}, p_{MH}, p_H, g_{\min})$ with $0 \le p_L < p_{ML} \le p_{MH} < p_H \le 1$ and $g_{\min} \in [0,1]$.

Remark 8 (Properties of the piecewise linear g). The function g defined in (B.1) satisfies: (i) g is continuous on $[0,1]$; (ii) g is non-decreasing on $[0, p_{MH}]$ and non-increasing on $[p_{MH}, 1]$; (iii) $g(p) = 0$ for $p < p_L$ and $g(p) \ge g_{\min}$ for all $p \ge p_H$; on the ascending ramp $[p_L, p_{ML})$, $g(p)$ can be below $g_{\min}$ when $g_{\min} > 0$; (iv) g is Lipschitz continuous, so expectations involving it are well-behaved. The five-parameter form is an implementation choice, not part of the contribution; any bounded, measurable weight would serve, and the parameters are chosen for their direct operational reading (above) rather than for approximation-theoretic reasons.

Remark 9 (Welfare-risk frontier). Each choice of g gives a pair of operator welfare and risk exposure. Observation 1 places the neutral weight $g \equiv 1$ at the highest-welfare point of the resulting curve, and it does not make the curve monotone in any parameter of g.

Remark 10 (When g could offset an estimation error). Observation 1 concerns oracle-optimal rules. If the true effect factors as $\tau(x) = \tau_0(x)\, m(R(x))$ and the effect model cannot capture the moderation m, a weight $g = m$ could improve an estimated rule.

This requires an operator-supplied g that mirrors a structural interaction no estimated component recovers, and the simulated misspecification (dropped features in the effect model) is not of that kind. The regime is a theoretical possibility, not a claim about the simulated data.

Numerical example

Example 1 (Illustrative targeting)**.** Eight hypothetical players have known treatment effects, operator values and churn risks (Table B.1). The values are abstract scores chosen so that three scoring rules select visibly different sets, and they do not respect the bound $\tau \leq p$ that a strict binary-retention reading would impose. We use $s_i = \tau_i V_i \, g(p_i; \theta_0)$ with $\theta_0 = (0.05, 0.40, 0.60, 0.90, 0.20)$ and a budget of 3 of 8 players.

Table B.1: Numerical example with eight hypothetical players: treatment effect τ (a probability), operator value V, churn risk p, preference weight $g(p)$, score s, and the targeting rank under three scoring rules. Budget: 3 players.

Player	τ	V	p	$g(p)$	s	Rank: τ	Rank: τV	Rank: $\tau V g$
A	0.35	200	0.03	0.00	0.0	1	4	7
B	0.30	500	0.25	0.57	85.7	2	2	3
C	0.20	800	0.50	1.00	160.0	3	1	1
D	0.15	600	0.45	1.00	90.0	4	3	2
E	0.10	400	0.80	0.47	18.7	5	6	4
F	0.05	950	0.10	0.14	6.8	7	5	5
G	−0.10	900	0.50	1.00	−90.0	8	8	8
H	0.08	300	0.95	0.20	4.8	6	7	6

Ranking on τ alone selects A, B and C. Player A has the largest effect but a churn risk of 0.03, so a bonus is wasted on a player who would stay anyway. Ranking on τV selects C, B and D, because A's low value of 200 drops A to fourth. The full score selects C, D and B: A is vetoed by $g = 0$, E is down-weighted for high churn, and C and D sit on the plateau. Player G is made worse off by treatment, and all three rules exclude G. None of the three rules considers harm, which is the subject of Section 5.

Welfare-risk frontier in the simulation

Table B.2 and Figure B.2 report operator welfare and risk exposure across fixed configurations of g, the calibrated configuration and an inverted g, at $\kappa = 0.30$ with welfare scored under $g \equiv 1$. Every fixed non-identity configuration sits below the neutral weight in welfare, consistent with, though not implied by, Observation 1, which covers oracle rules. The inverted g, which puts high weight at extreme churn and low weight in the moderate zone, earns 13,934 at risk exposure 0.145 and is also below the neutral weight. The calibrated row is the AIPW-calibrated rule, which does not reproduce the identity and sits below it (Appendix C). Risk exposure varies far less across configurations than welfare does, so in this simulation the dial on g is mostly a welfare cost and only weakly reshapes who is treated. A population in which churn risk clusters more sharply in the treated tail would show wider variation in exposure.

Table B.2: Welfare and risk exposure under different configurations of g ($\kappa = 0.30$, welfare scored under $g \equiv 1$). Labels show $(p_L/p_{ML}/p_{MH})$, with $p_H = 0.90$ and $g_{\min} = 0.20$ unless stated. Calibrated is the AIPW-calibrated configuration. Risk exposure is mean churn probability among the treated.

Configuration	Welfare	Risk exposure
No reg (g=1)	16,446	0.136
Wide (0.03/0.50/0.70)	12,599	0.148
Default (0.05/0.40/0.60)	13,646	0.146
Narrow (0.10/0.30/0.50)	14,935	0.145
Aggressive (0.15/0.35/0.50/$g_{\min}$=0.05)	13,738	0.149
Calibrated	15,566	0.142
Inverted g	13,934	0.145

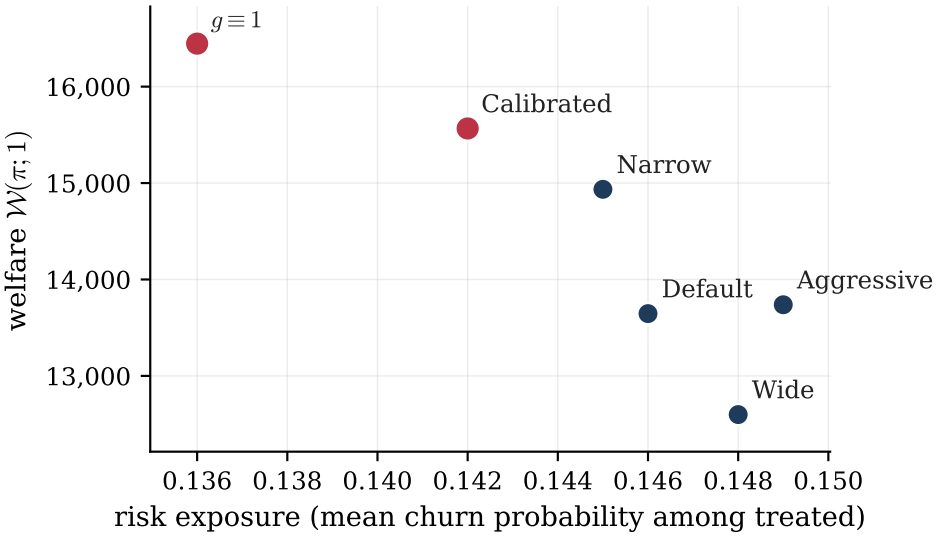


Figure B.2: Welfare-risk frontier: each fixed configuration of g sits below the neutral weight $g \equiv 1$ in welfare. This is consistent with, though not implied by, Observation 1, which covers oracle rules.

Appendix C. Calibration of the preference weight

This appendix describes how the preference weight is calibrated in the simulation and what the calibration shows. Each replication splits its training set 60/40 into an estimation fold, which fits the component models, and a validation fold, which scores candidate configurations of θ. The candidate set has 22 configurations, including a near-identity configuration ($p_L = 0$, $p_{ML} = 0.01$, $p_{MH} = 0.99$, $g_{\min} = 1$) that makes g constant on $[0.01, 0.99]$. Because $\hat{p} \geq 0.01$, this configuration is equivalent to the identity, and the two select the same set. Each candidate rule treats the top κ share of the validation fold by $\hat{\tau}\hat{V}\, g(\hat{p}; \theta)$, and the objective scores the selected set. Ties go to the lowest grid index, and no oracle information enters the selection.

The plug-in objective sums $\hat{\tau}\hat{V}$ over the selected set. The identity candidate selects exactly the set that maximises this sum, so the plug-in objective picks the identity, or a candidate with the same plug-in value, by construction (Observation 1). It selects the identity in 200 of 200 replications with default learners, where its welfare difference from the identity is exactly zero and has no p-value. With tuned learners it selects the identity in 198 replications. The tuned learners produce coarse scores with many

ties at the threshold, so in the remaining replications another candidate can match the identity's plug-in welfare up to floating-point rounding while selecting a different tied set; the resulting welfare difference is not distinguishable from zero (Table C.1). These rows carry no evidence about whether g helps.

The AIPW objective sums $\hat{\Gamma}_i \hat{V}_i$ over the selected set, where $\hat{\Gamma}_i$ is the cross-fitted doubly robust score of the effect under the known propensity 0.5. Nothing forces it to pick the identity, and it selects the identity in 13 of 200 replications (6.5%) with default learners and 13 (6.5%) with tuned learners. Table C.1 shows that AIPW calibration costs operator welfare against the identity: −881 [−1,217, −544], $d = -0.37$, or −5.4% [−7.4%, −3.3%], with default learners, and −373 [−667, −78], $d = -0.18$, with tuned learners. Regret against the feasible oracle is 17,231 for AIPW calibration and 16,350 for the plug-in rule with default learners. One reading, which we did not test, is that the validation fold is small and the AIPW objective noisy, so the maximum over 22 candidates tends to pick a configuration that looks good on that fold by chance. No configuration in the grid beats the identity on true welfare in this design, and the fixed configurations of Appendix B are all below it as well.

Table C.1: Calibration of the preference weight ($\kappa = 0.30$, 200 replications). Identity selected gives the share and number of replications in which calibration selected the identity configuration. Calibrated minus identity is the welfare difference with a 95% confidence interval, and regret is against the feasible oracle with a 95% confidence interval. † The plug-in objective selects the identity by construction, so its difference is exactly zero and it has no p-value.

		Identity selected				
Objective	Learners	Rate	Count	Calibrated minus identity	Holm p	Regret vs feasible oracle
AIPW	default	6.5%	13/200	−881 [−1,217, −544]	<0.001	17,231 [16,669, 17,792]
Plug-in (degenerate†)	default	100.0%	200/200	+0 [0, 0]	–	16,350 [15,917, 16,784]
AIPW	tuned	6.5%	13/200	−373 [−667, −78]	0.0272	16,228 [15,347, 17,109]
Plug-in	tuned	99.0%	198/200	+4 [−3, 10]	0.274	15,852 [14,998, 16,706]

Appendix D. Harm sweep welfare and the tight budget

Table D.1 gives operator welfare for every cell of the harm sweep of Section 6.2. The harm-blind rule's welfare does not depend on c_{harm} or σ_h because it never reads harm and welfare counts only $\tau_i V_i$. Welfare differences between the three estimated rules are largest at $c_{\text{harm}} = 1$. The population mean harm falls from 0.458 at $c_{\text{harm}} = -1$ to 0.421 at $c_{\text{harm}} = 1$, and the share of players above the soft weight's lower threshold 0.6 falls from 29.0% to 20.8%. The mean drifts although the variance of the harm logit is fixed, because z_{spend} is right-skewed. Because the share of players the soft weight can down-weight differs across rows, welfare differences across c_{harm} mix the effect of alignment with the effect of this shift in the harm distribution. The soft weight also beats harm-blind ranking at $c_{\text{harm}} = -1$ at every σ_h (Table D.1), which fits the regularisation reading of Section 6.3:

where harm falls with value, a high $\hat{H}$ flags low value, and the weight trims the lower tail of $\hat{\tau}\hat{V}$ instead of the upper one. Table D.2 repeats the exposure sweep of Table 2 at $\kappa = 0.10$, where the harm-blind gap at $c_{\text{harm}} = 1$ is +0.152 and both harm-aware rules stay above the population mean.

Table D.1: Operator welfare in the harm sweep ($\kappa = 0.30$). Rules and component sources are those of Table 2.

		Estimated components			Feasible-oracle components		
c_{harm}	σ_h	Harm-blind	Soft weight	Hard cap	Harm-blind	Soft weight	Hard cap
−1.0	0.0	16,446	16,915	16,617	32,797	32,496	32,707
−1.0	0.5	16,446	16,839	16,600	32,797	32,215	32,533
−1.0	1.0	16,446	16,615	16,487	32,797	31,459	32,086
−0.5	0.0	16,446	16,381	16,355	32,797	31,455	32,079
−0.5	0.5	16,446	16,363	16,318	32,797	31,275	31,930
−0.5	1.0	16,446	16,297	16,193	32,797	30,772	31,566
0.0	0.0	16,446	16,387	16,189	32,797	31,094	31,851
0.0	0.5	16,446	16,377	16,156	32,797	30,895	31,704
0.0	1.0	16,446	16,420	16,033	32,797	30,483	31,359
0.5	0.0	16,446	17,537	16,058	32,797	31,156	31,901
0.5	0.5	16,446	17,516	16,015	32,797	30,961	31,771
0.5	1.0	16,446	17,397	15,893	32,797	30,454	31,380
1.0	0.0	16,446	19,045	15,701	32,797	31,986	32,361
1.0	0.5	16,446	18,897	15,703	32,797	31,663	32,185
1.0	1.0	16,446	18,582	15,669	32,797	30,909	31,719

Appendix E. Additional scenarios and tuning details

Scenarios

Table E.1 reports the comparison of CATE × LTV with CATE only, and a few other contrasts, in each additional scenario at $\kappa = 0.30$ unless stated. The first difference column uses the default T-learner and the last two columns repeat the comparison with the tuned T-learner, which was tuned on the main design only and reused unchanged. The scenarios are the main design; a null design with no effect-churn interaction ($m(p) \equiv 1$); correlated covariates drawn through a Gaussian copula; a misspecified effect model that drops login frequency while the value and churn models keep all features; an asymmetric moderation $m(p) \propto p(1-p)^3$ that peaks near $p = 0.25$; a correlation sweep with $V \leftarrow V + \gamma \cdot 180\, z(\tau_{\text{base}})$; a heavy-tailed value V with a multiplicative log-normal term, clipped at 100,000; and an adversarial design in which the effect also depends on bonus history, which does not enter V, and the effect model drops it.

Misspecified effect model and the adversarial case. With login frequency dropped from the effect model, CATE × LTV beats CATE only by +287 [170, 404], $d = 0.34$, with the default T-learner, because login frequency also enters value and the value model therefore carries part of the lost signal. With the tuned T-learner the sign reverses, to −2,265 [−2,666, −1,865], $d = -0.79$, so the rescue depends on the learner. In the adversarial design the value model cannot recover the dropped signal, and value weighting does not

Table D.2: Harm exposure of the treated minus the population mean at the tight budget ($\kappa = 0.10$, 95% confidence intervals in brackets). The layout follows Table 2.

c_{harm}	σ_h	Harm-blind	Soft weight	Hard cap
Panel A: estimated components				
−1.0	0.0	−0.142 [−0.148, −0.137]	−0.160 [−0.165, −0.155]	−0.160 [−0.165, −0.156]
−1.0	0.5	−0.142 [−0.148, −0.137]	−0.163 [−0.168, −0.158]	−0.165 [−0.170, −0.161]
−1.0	1.0	−0.142 [−0.148, −0.137]	−0.165 [−0.170, −0.160]	−0.171 [−0.176, −0.166]
−0.5	0.0	−0.074 [−0.078, −0.071]	−0.130 [−0.133, −0.127]	−0.131 [−0.134, −0.128]
−0.5	0.5	−0.074 [−0.078, −0.071]	−0.131 [−0.133, −0.128]	−0.132 [−0.135, −0.129]
−0.5	1.0	−0.074 [−0.078, −0.071]	−0.130 [−0.132, −0.127]	−0.132 [−0.135, −0.130]
0.0	0.0	−0.001 [−0.003, 0.001]	−0.084 [−0.086, −0.082]	−0.085 [−0.087, −0.083]
0.0	0.5	−0.001 [−0.003, 0.001]	−0.083 [−0.085, −0.081]	−0.084 [−0.086, −0.082]
0.0	1.0	−0.001 [−0.003, 0.001]	−0.081 [−0.082, −0.079]	−0.082 [−0.084, −0.080]
0.5	0.0	+0.077 [0.074, 0.081]	−0.042 [−0.045, −0.040]	−0.043 [−0.046, −0.041]
0.5	0.5	+0.077 [0.074, 0.081]	−0.042 [−0.044, −0.039]	−0.040 [−0.042, −0.037]
0.5	1.0	+0.077 [0.074, 0.081]	−0.035 [−0.038, −0.033]	−0.031 [−0.033, −0.028]
1.0	0.0	+0.152 [0.146, 0.158]	+0.018 [0.015, 0.021]	+0.017 [0.014, 0.021]
1.0	0.5	+0.152 [0.146, 0.158]	+0.019 [0.016, 0.023]	+0.031 [0.028, 0.035]
1.0	1.0	+0.152 [0.146, 0.158]	+0.023 [0.020, 0.027]	+0.047 [0.043, 0.051]
Panel B: feasible-oracle components				
−1.0	0.0	−0.066 [−0.067, −0.065]	−0.085 [−0.086, −0.084]	−0.086 [−0.087, −0.085]
−1.0	0.5	−0.066 [−0.067, −0.065]	−0.090 [−0.091, −0.088]	−0.093 [−0.094, −0.092]
−1.0	1.0	−0.066 [−0.067, −0.065]	−0.091 [−0.092, −0.090]	−0.097 [−0.098, −0.096]
−0.5	0.0	−0.022 [−0.024, −0.020]	−0.108 [−0.109, −0.106]	−0.108 [−0.110, −0.107]
−0.5	0.5	−0.022 [−0.024, −0.020]	−0.106 [−0.107, −0.104]	−0.107 [−0.109, −0.106]
−0.5	1.0	−0.022 [−0.024, −0.020]	−0.099 [−0.101, −0.098]	−0.103 [−0.105, −0.102]
0.0	0.0	+0.001 [−0.001, 0.003]	−0.107 [−0.109, −0.106]	−0.108 [−0.110, −0.107]
0.0	0.5	+0.001 [−0.001, 0.003]	−0.104 [−0.106, −0.103]	−0.106 [−0.107, −0.104]
0.0	1.0	+0.001 [−0.001, 0.003]	−0.096 [−0.097, −0.094]	−0.099 [−0.101, −0.098]
0.5	0.0	+0.021 [0.019, 0.023]	−0.083 [−0.084, −0.082]	−0.084 [−0.085, −0.082]
0.5	0.5	+0.021 [0.019, 0.023]	−0.080 [−0.081, −0.078]	−0.081 [−0.082, −0.080]
0.5	1.0	+0.021 [0.019, 0.023]	−0.071 [−0.073, −0.070]	−0.073 [−0.075, −0.072]
1.0	0.0	+0.052 [0.051, 0.054]	+0.001 [0.000, 0.002]	+0.000 [−0.001, 0.001]
1.0	0.5	+0.052 [0.051, 0.054]	−0.001 [−0.002, 0.000]	+0.000 [−0.001, 0.001]
1.0	1.0	+0.052 [0.051, 0.054]	+0.001 [0.000, 0.002]	+0.002 [0.001, 0.004]

help: at $\kappa = 0.30$ the difference is −133 [−302, 36], and at $\kappa = 0.10$ it is −1,262 [−1,426, −1,098], $d = -1.07$, or −24.3% [−27.7%, −21.0%]. The tuned T-learner never splits on bonus history, so dropping it changes nothing for that learner. The tuned adversarial rows therefore compare value weighting under a learner that ignores the adversarial feature, not the effect of dropping it.

Null design. With no effect-churn interaction, CATE × LTV beats CATE only by +5,157 [4,704, 5,610], $d = 1.59$, with the default T-learner. With the tuned T-learner the difference is +191 [−1,988, 2,370], whose interval includes zero, so the null-design advantage of value weighting holds for the default T-learner only.

Feature dropout like for like. Dropping login frequency from the effect model costs CATE only +9,934 [9,554, 10,314], a loss of +59.7%, and costs the S-learner +15,441 [14,704, 16,178], a loss of +63.6%. Dropping bonus history in the adversarial design costs CATE only +13,636 [13,207, 14,066]. The S-learner is not more robust to losing an informative

Table E.1: Additional scenarios (200 replications, $\kappa = 0.30$ unless named). Panel A gives CATE × LTV minus CATE only in each scenario; panel B gives the contrast named in the row, first named minus second. Each difference is shown above its 95% confidence interval, d is Cohen's d, and Family is the size of the scenario's Holm family. For the like-for-like rows, a positive value is a loss from dropping the feature. The last two columns repeat the comparison with the tuned T-learner where one exists; pp is percentage points.

	Default T-learner				Tuned T-learner	
Scenario or contrast	Difference [95% CI]	d	Holm p	Family	Difference [95% CI]	Holm p
Panel A: CATE × LTV minus CATE only, by scenario						
Main design	−187 [−319, −55]	−0.20	0.0170	24	−1,819 [−2,312, −1,326]	<0.001
Null design	+5,157 [4,704, 5,610]	1.59	<0.001	24	+191 [−1,988, 2,370]	0.863
Correlated covariates	−678 [−800, −556]	−0.77	<0.001	24	−4,215 [−4,795, −3,636]	<0.001
Misspecified effect model	+287 [170, 404]	0.34	<0.001	24	−2,265 [−2,666, −1,865]	<0.001
Asymmetric moderation	+398 [152, 644]	0.23	0.0016	8	−2,716 [−3,750, −1,682]	<0.001
Correlation sweep, $\gamma = -1$	−123 [−258, 12]	−0.13	0.0742	18	−946 [−1,308, −584]	<0.001
Correlation sweep, $\gamma = 0$	−329 [−464, −194]	−0.34	<0.001	18	−1,664 [−2,098, −1,231]	<0.001
Correlation sweep, $\gamma = 1$	−504 [−647, −362]	−0.49	<0.001	18	−2,390 [−2,930, −1,850]	<0.001
Heavy-tailed value	−733 [−1,041, −425]	−0.33	<0.001	6	−2,910 [−3,698, −2,123]	<0.001
Adversarial design, $\kappa = 0.30$	−133 [−302, 36]	−0.11	0.319	16	−1,739 [−2,270, −1,208]	<0.001
Adversarial design, $\kappa = 0.10$	−1,262 [−1,426, −1,098]	−1.07	<0.001	16	−2,994 [−3,320, −2,669]	<0.001
Panel B: other contrasts						
CATE × LTV, nominal minus clip-adjusted welfare	+787 [743, 832]	2.47	<0.001	17	–	–
CATE × LTV, well-specified minus proxy value model	−532 [−573, −491]	−1.81	<0.001	2	–	–
Churn-weighted, AIPW-calibrated minus identity g	−881 [−1,217, −544]	−0.37	<0.001	3	–	–
CATE only, all features minus login frequency dropped	+9,934 [9,554, 10,314]	3.65	<0.001	5	–	–
CATE only, all features minus bonus history dropped (adversarial design)	+13,636 [13,207, 14,066]	4.43	<0.001	5	–	–
Effect share of the score-gap bound, login dropped minus all features (T-learner, plug-in reference)	+1.5 pp [1.4, 1.7] pp	1.71	<0.001	6	–	–

feature; its advantage is that it uses a given feature set more effectively, and comparing it with a rule whose effect model lost the feature is not like for like, which is why learner choice is a separate question. An observational diagnostic bounds the mean absolute score gap over the misranked set and splits the bound into an effect term and a value term in one fixed telescoping order. It is not a regret share, because only the Shapley split of Remark 5 decomposes regret. The per-replication mean effect share is 93.1% [92.9%, 93.2%] for the full-feature T-learner against a held-out plug-in reference, and the share rises by +1.5*pp* [1.4*pp*, 1.7*pp*] when login is dropped. Against the feasible and hindsight oracles the same full-feature share is 76.9% [76.7%, 77.1%] and 79.5% [79.3%, 79.6%], a score-error heuristic that is lower than the Shapley share of Section 6.6.

Other scenarios. Under the asymmetric moderation CATE × LTV beats CATE only by +398 [152, 644], whereas on the main design the sign is negative (−187), and under the tuned T-learner the asymmetric gain reverses to −2,716 [−3,750, −1,682], so whether value weighting helps at a given budget depends on a moderation shape the operator cannot observe in advance. In the correlation sweep the difference is −123 [−258, 12], −329 [−464, −194] and −504 [−647, −362] for $\gamma = -1$, 0 and 1, with the same sign and a larger magnitude under the tuned T-learner. With a heavy-tailed value the difference is −733 [−1,041, −425], or −2.7% of CATE only. Nominal minus clip-adjusted CATE × LTV welfare is +787 [743, 832], $d = 2.47$, so reported welfare is nominal and overstates realised lift. The overstatement is larger than the headline difference it qualifies, but it does not change that difference's sign. CATE × LTV overstates its welfare by +197 [180, 214] more than CATE only (Holm $p < 0.001$), and on clip-adjusted welfare CATE only earns 16,044 against 15,659 for CATE × LTV, a difference of −385 against −187 on nominal welfare. The clip-adjusted difference is a difference of replication means, for which no paired interval is stored. Replacing the proxy value model with the well-specified one changes CATE × LTV by −532 [−573, −491], the proxy-beats-well-specified result of Section 6.6.

Learner tuning

Each learner is tuned once, on the main design, by mean cross-fitted R-loss over 5 pilot replications, and the tuned parameters are then fixed for every study. The grid has 80 configurations: depth 1 to 4, 25, 50, 100 or 300 trees, and learning rate 0.005, 0.01, 0.02, 0.05 or 0.1. Within a pilot replication the training split is cross-fitted into two folds, the learner is fitted on one and scored on the other by $\text{mean}\Big((Y - \hat{m} - (W - 0.5)\,\hat{\tau})^2\Big)$ with $\hat{m}$ a fixed gradient-boosted estimate of $\mathbb{E}[Y \mid X]$. The pilot seeds are disjoint from the evaluation seeds. The chosen configurations are depth 1, 25 trees and rate 0.020 for the T-learner; depth 2, 100 trees and rate 0.010 for the X-learner; depth 2, 300 trees and rate 0.100 for the S-learner; and depth 1, 100 trees and rate 0.005 for the DR-learner. The T-learner chooses the smallest depth and tree count, the DR-learner the smallest depth and learning rate, and the S-learner the largest tree count and learning rate, so all three tuned fits may be limited by the grid. For T and DR the edge is toward more shrinkage, which strengthens the attenuation reading, although a wider grid could change the magnitudes. The chosen R-loss is 0.21258 (T), 0.21255 (X), 0.21253 (S) and 0.21257 (DR), a spread across learners of 0.00005, while the worst-to-best range across the grid is 0.01573 (T), 0.00803 (X), 0.00021 (S) and 0.02108 (DR). The S-learner's loss is almost flat across its grid, so its tuned configuration is weakly identified. R-loss measures how well the estimate calibrates against residualised outcomes and does not measure how well it ranks players, so a lower R-loss need not mean better targeting.

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